

Identification Quality Over Finding Direction: Evidence from Government Intervention Research in Top Finance Journals*

[Author Name][†] [Co-Author Name][‡]

This version: June 3, 2026

[Working paper — please do not cite without permission]

ABSTRACT

We examine whether top finance journals exhibit publication bias in the government-intervention literature. Using a two-sample design—157 NBER working papers and 134 top-journal articles spanning 2000–2024—we classify the direction of each paper’s primary empirical finding and compare cross-sectional publication rates. We find no significant difference in publication rates across finding directions: positive-finding papers appear in the *Journal of Finance*, *Review of Financial Studies*, or *Journal of Financial Economics* at 45.6%, negative-finding papers at 46.1%, and null-finding papers at 46.7% ($\chi^2 = 0.017$, $p = 0.992$). A binary directional probit coefficient is -0.020 (s.e. 0.182, $p = 0.91$), and the three-way direction indicators are uniformly insignificant across specifications. The only robust predictor of top-journal publication is the use of a quasi-experimental identification strategy (RDD, IV, DiD, or event study): the quasi-experimental dummy carries a coefficient of 0.467^{***} (s.e. 0.157, $p = 0.003$), implying an approximately 18 percentage-point publication premium that is robust to all controls. A complementary analysis of within-paper test

*We thank seminar participants at [XX] for helpful comments. We are grateful to [XX] for outstanding research assistance. All data and code are available at [https://github.com/\[REPOSITORY\]](https://github.com/[REPOSITORY]). Correspondence: [author@university.edu].

[†][Department of Finance, University of XX.]

[‡][Department of Economics, University of YY.]

statistic distributions reveals significant threshold bunching in the published record (Brodeur et al. 2016 caliper ratio = 1.20, $p = 0.002$), but this bunching is entirely absent from the pre-publication NBER benchmark (ratio = 1.01, $p = 0.849$). The contrast is sharpest for negative-finding papers: published caliper ratio 1.52 ($p < 0.001$) versus NBER ratio 1.00 ($p = 0.964$). Because the overall significance rates are nearly identical across samples (57.1% vs. 55.5%), the bunching is not explained by differences in statistical power; it is concentrated in the published record and absent from pre-publication counterparts, consistent with revision-stage pressure. We interpret the joint pattern as evidence that editorial selection in top finance journals is directionally neutral, but that the revision process creates within-paper threshold pressure, particularly for anti-intervention findings.

JEL classification: G18, G28, C80, B40.

Keywords: Publication bias; Government intervention; Finance journals; Identification quality; Caliper test; Working papers

1. Introduction

Academic research on government intervention in financial markets occupies a privileged position in public policy. The Federal Reserve Board, the Securities and Exchange Commission, and the Basel Committee on Banking Supervision routinely incorporate peer-reviewed finance research into regulatory impact assessments, rule-makings, and policy speeches. The Dodd-Frank Wall Street Reform and Consumer Protection Act alone generated an estimated 400 new regulatory mandates, many of which were designed in explicit reference to academic evidence on the effects of capital requirements, disclosure rules, and consumer financial protection. The credibility of this evidence base is therefore a first-order policy question: if the academic record systematically misrepresents which interventions have measurable effects—and which do not—regulators face a distorted picture that can generate welfare-reducing policy choices.

Publication bias offers a mechanism by which such distortion arises without any deliberate misconduct. When researchers are reluctant to write up null results, when journals are more likely to accept papers with statistically significant findings, or when both effects operate simultaneously, the published literature understates the true prevalence of null or inconclusive evidence (Franco et al., 2014). In a domain such as government intervention research, where policy decisions depend on whether a regulation has a measurable causal effect, suppression of null results generates the false impression that most interventions work—or, more precisely, that most interventions have discernible effects in one direction or the other.

This paper studies publication bias in a domain the existing literature has not examined: the finance literature’s treatment of government intervention. Prior work on publication bias in finance has focused on the cross-section of asset pricing anomalies, documenting that factor-zoo proliferation reflects selection for high t-statistics (Harvey et al., 2016; Hou et al., 2020; Jensen et al., 2023). A parallel economics literature (Brodeur et al., 2016; Andrews and Kasy, 2019) infers bias from the shape of published p-value distributions,

finding excess mass just below conventional significance thresholds. Neither strand addresses the policy-intervention literature, nor do they distinguish two conceptually distinct forms of bias with different welfare implications. *Significance bias*—directional findings over-published relative to null findings—makes every intervention appear detectable. *Directional bias*—pro-intervention findings over-published relative to anti-intervention findings—additionally skews the regulatory record toward a particular conclusion about the sign of policy effects. We design our study to test for both.

Our research design combines two complementary samples. The *published paper* sample consists of all in-scope articles published in the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics* from 2000 to 2024. The *working paper* sample is drawn from NBER working papers in five programs: Corporate Finance (CF), Asset Pricing (AP), Monetary Economics (ME), Economic Fluctuations and Growth (EFG), and Public Economics (PE), covering papers from 2000 to 2024. A key methodological advantage of this two-sample design is that we observe actual pre-publication findings rather than merely the shape of the published distribution: the gap between pre- and post-publication finding rates is directly observable rather than inferred from p-value bunching, though the cross-sectional design cannot rule out that it reflects differences between the two populations rather than the editorial filter alone. Together, these samples allow us to compare the full distribution of findings across publication states—not merely the subset that appears in top journals.

We define a paper as in-scope if its primary research question is the causal effect of a specific government law, regulation, or public policy on financial markets, firms, or investors. Identifying in-scope papers in a corpus of more than 40,000 documents would be infeasible by hand. We therefore develop a two-pass automated pipeline: a keyword filter that screens for government intervention terminology, followed by a large language model (LLM) classifier that reads each flagged abstract and returns a binary in-scope decision with a structured justification. For each in-scope paper we then code the *direction* of the primary finding as

positive (+1, the intervention was beneficial), negative (−1, harmful), or null/mixed (0).

Our primary finding is the *absence* of a significance-sorting pattern in the published record. Publication rates are nearly identical across all three direction categories: positive-finding papers appear in top journals at 45.6%, negative-finding papers at 46.1%, and null-finding papers at 46.7%. A χ^2 test of homogeneity cannot reject equality ($\chi^2 = 0.017$, $p = 0.992$), and a two-sample z -test for equality of the positive and negative rates likewise fails to reject ($z = -0.07$, $p = 0.94$).

Probit models confirm the absence of a direction premium. The binary directional coefficient is -0.020 (s.e. 0.182 , $p = 0.91$), and the three-way direction indicators are -0.026 (positive, $p = 0.90$) and -0.014 (negative, $p = 0.94$).¹ A Wald test for equality of the two direction coefficients gives $\chi^2 = 0.005$ ($p = 0.944$). These null results are robust across all six specifications.

The one robust predictor of top-journal publication is quasi-experimental identification quality. The *QuasiExp* dummy carries a coefficient of 0.467^{***} (s.e. 0.157 , $p = 0.003$) in Column (6), implying an 18 percentage-point publication premium for papers using RDD, IV, DiD, or event-study designs. This identification-quality premium survives all controls, whereas the direction coefficients remain statistically indistinguishable from zero. The evidence suggests that editorial selection in this literature rewards methodological rigor, not finding direction.

The absence of a direction–publication association is uniform across journals. No journal shows a statistically significant direction coefficient: the *Journal of Finance* yields $\hat{\beta}_{\text{pos}} = 0.393$ ($p = 0.27$) and $\hat{\beta}_{\text{neg}} = 0.329$ ($p = 0.35$). The *Review of Financial Studies* yields $\hat{\beta}_{\text{pos}} = -0.036$ ($p = 0.90$) and $\hat{\beta}_{\text{neg}} = 0.252$ ($p = 0.35$). The *Journal of Financial Economics* yields $\hat{\beta}_{\text{pos}} = -0.136$ ($p = 0.57$) and $\hat{\beta}_{\text{neg}} = -0.305$ ($p = 0.20$), both insignificant. Cross-journal differences are not statistically distinguishable at available sample sizes (see

¹The baseline implied publication probability at the bivariate intercept is $\Phi(-0.084) \approx 46.7\%$, nearly identical to all three direction-specific publication rates. Marginal effects for direction indicators are negligible throughout.

Section 5).

A complementary analysis of the *within-paper* distribution of test statistics reveals a distinct pattern and, crucially, identifies where in the research pipeline it originates. Applying the Brodeur et al. (2016) caliper test to 7,761 $|z|$ -statistics from 121 published papers, we find significant full-sample threshold bunching (caliper ratio = 1.20, $p = 0.002$), concentrated in *Journal of Financial Economics* (caliper ratio = 1.49, $p < 0.001$) and strikingly asymmetric by direction: *negative-finding papers* show a caliper ratio of 1.52 ($p < 0.001$) while positive-finding papers show no bunching (ratio = 0.94, $p = 0.565$). We then apply the identical extraction pipeline to the full-text PDFs of the 157 in-scope NBER working papers, obtaining 8,351 pre-publication test statistics from 125 papers. The NBER caliper ratio is 1.01 ($p = 0.849$) overall, and 1.00 ($p = 0.964$) for negative-finding papers—essentially exactly one. The overall significance rate is nearly identical across samples (published: 57.1%; NBER: 55.5%), ruling out differences in statistical power. The bunching does not appear to be a property of the underlying research designs in this domain; it is concentrated in the published record and absent from pre-publication counterparts, consistent with revision-stage pressure—though the cross-sectional design cannot rule out that it partly reflects compositional differences between papers that circulate through NBER versus those that reach top journals.

Sub-period estimates across three regulatory regimes (2000–2007, 2008–2015, 2016–2024) are uniformly insignificant, with all direction coefficients well below conventional significance thresholds. Sub-period sample sizes ($N = 50, 95, 146$) are small and these results should be interpreted as descriptive, not confirmatory.

This paper contributes to three strands of the literature. It provides direct evidence on the *absence* of direction-based publication bias in the government-intervention finance literature, a finding that stands in contrast to the widespread significance-sorting documented in the broader economics literature. The result suggests that finance journals in this domain select primarily on methodological rigor (quasi-experimental identification) rather than on

whether papers find that a policy has a measurable effect.

The paper also contributes methodologically to the literature on identifying publication bias. Unlike approaches that infer bias from the internal shape of published p-value distributions, our NBER working paper baseline allows us to observe actual pre-publication findings and measure the cross-sectional sorting gap. The AFA conference comparison provides a consistency check: at the pre-journal presentation stage, directional papers carry no statistically significant association with eventual publication ($\hat{\beta} = 0.327$, $p = 0.24$), consistent with the NBER baseline result and with the absence of a publication-stage direction premium. Both pre-publication baselines (NBER and AFA) support the same conclusion: no evidence of significance-sorting—though supply-side submission decisions at that stage cannot be ruled out.

Finally, the paper introduces a scalable LLM-assisted pipeline for classifying and direction-coding large economics paper corpora. Throughout, we are explicit that our design identifies a cross-sectional association between finding direction and publication status; it cannot distinguish editorial selection from author self-selection in submission or specification search.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes data construction and the classification pipeline. Section 4 presents the empirical strategy. Section 5 reports main results and economic magnitudes. Section 6 presents robustness checks. Section 7 discusses mechanisms, policy implications, comparisons to other domains, and limitations. Section 8 concludes.

2. Related Literature

2.1 Publication Bias in Asset Pricing

The closest prior work on publication bias in finance examines the cross-section of expected returns. Harvey et al. (2016) survey hundreds of return predictors published in top finance journals and argue that the conventional $t > 2$ threshold is inadequate given the scale of

multiple testing in the factor zoo. Hou et al. (2020) attempt to replicate 452 published anomalies and find that 65% fail under standard portfolio construction choices, consistent with upward-biased published effect sizes. Jensen et al. (2023) document that most factors replicate out-of-sample but that the factor space is highly redundant, implying substantial duplication in published claims. Harvey and Liu (2021) develop a structural model to infer the pre-publication distribution of test statistics—the full iceberg—from the published distribution alone, which is only the tip. They estimate the share of true factors and the intensity of specification search as latent parameters; identification is challenging because the pre-publication distribution is unobserved. Chen (2021) takes a complementary approach, running thought experiments to bound how much p-hacking would be needed to generate the observed right tail of published t-statistics if all 300+ asset-pricing factors were spurious. He finds the implied search effort is implausibly large—if 10,000 researchers generated one factor per minute it would take millions of years—concluding that most published factors reflect genuine economic content rather than data-mining. Both papers study the *level* of published test statistics in asset pricing.

This strand focuses entirely on *significance-level* sorting in the asset-pricing factor zoo. All papers in it face the identification problem that the pre-publication distribution is unobserved and must be inferred from a structural model (Harvey & Liu) or bounded by thought experiment (Chen). Our paper sidesteps this problem by directly observing the pre-publication distribution through NBER working paper full texts, converting an inferential exercise into a direct empirical comparison.

The two papers are also studying a different part of the t-statistic distribution. Chen’s bound concerns the *far* right tail ($|t| > 4$, $|t| > 6$), which is too heavy to be explained by p-hacking alone. Our caliper test focuses on the narrow window *around* the 5% threshold ($|z| \in (1.645, 2.275)$), where selective reporting of marginally significant specifications creates excess mass just *below* the 1.96 boundary. Both phenomena can coexist: genuine economic effects produce the fat tail; revision-stage pressure produces excess mass in the borderline

10%-significant zone. Indeed, our findings are entirely consistent with Chen’s conclusion that most published effects are real—we find no evidence that publication *directionally* sorts pro- versus anti-intervention findings, and our caliper evidence points to within-paper revision pressure rather than outright fabrication.

Our paper also differs on the question asked. Factor-return sign is not normatively coded, so these papers cannot test for *directional* bias—whether papers concluding an intervention is harmful are less likely to reach print than papers concluding it is beneficial. That question is economically meaningful only in a domain where finding direction has a clear policy interpretation, which is precisely the government-intervention literature we study.

2.2 Publication Bias in Economics

The economics literature on publication bias draws on a long tradition in statistics and medicine. Brodeur et al. (2016) analyze 50,000 hypothesis tests reported in the *American Economic Review*, the *Journal of Political Economy*, and the *Quarterly Journal of Economics*, finding excess mass just below the 5% significance threshold ($|z| \in (1.645, 1.96)$), consistent with selective reporting of marginally significant specifications. Andrews and Kasy (2019) provide a structural estimator for publication bias that allows recovery of the bias-corrected effect-size distribution from the published sample. Ioannidis et al. (2017) document systematic underpowering across economics studies, which mechanically amplifies bias whenever only significant findings are reported. Christensen and Miguel (2018) survey the broader reproducibility literature and document that publication bias, p-hacking, and HARKing (hypothesizing after results are known) are pervasive in empirical economics. These approaches infer bias from the internal shape of the published distribution—they do not observe pre-publication findings and therefore cannot attribute the sorting gap to editorial selection specifically. Our paper contributes domain-specific evidence for the finance-regulation literature using a pre-publication baseline that identifies the editorial stage directly.

2.3 Government Intervention in Financial Markets

Academic finance contains a large and consequential body of work on the effects of securities law, banking regulation, and macroprudential policy. Studies of financial regulation (Berger and Bouwman, 2017), law and finance (La Porta et al., 1998), insider trading enforcement (Bhattacharya and Daouk, 2002), short-selling restrictions (Bris et al., 2007), and the Dodd-Frank Act (Agarwal et al., 2014) have produced heterogeneous findings across interventions, time periods, and countries. Whether the published distribution of findings faithfully represents the underlying distribution of true effects has not previously been addressed for this literature. We fill this gap.

2.4 LLMs in Economics Research

Recent work demonstrates that large language models can accurately classify economic texts, extract structured data from unstructured documents, and replicate tasks that previously required extensive human coding (Ash and Hansen, 2023; Korinek, 2023). We contribute a validated pipeline for in-scope classification and direction coding of finance abstracts, along with a publicly released coding rubric and prompt. Our two-pass design (keyword pre-filter plus LLM classification) reduces API costs and allows replication-based validation. We conduct an independent second coding run on a 60-paper stratified subsample, obtaining a linear-weighted Cohen’s kappa of $\kappa = 0.674$, consistent with “substantial agreement” by the Landis and Koch (1977) classification, and addressing concerns about LLM reliability in structured classification tasks (Brodeur et al., 2023).

3. Data and Sample Construction

3.1 Published Papers

The published-paper sample consists of all articles appearing in the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics* from 2000 to 2024.

These three journals constitute the traditional top tier of academic finance and account for the majority of highly cited finance research on government intervention. We collect article metadata—title, authors, year, volume, issue, DOI, and abstract—for all articles in each journal over this period.

Journal articles are identified and retrieved as follows. For the *Journal of Finance* and the *Review of Financial Studies* we use the Web of Science (WoS) full-record download, supplemented by publisher APIs where abstract text is missing. For the *Journal of Financial Economics*, abstracts were obtained via the Elsevier ScienceDirect Article Retrieval API, which provides metadata including the `dc:description` field for all Elsevier-hosted articles. Table 1 summarizes the journal sources.

Table 1. Journal Sample

Journal	Abbrev.	ISSN	Publisher	Abstract source
<i>Journal of Finance</i>	JF	0022-1082	Wiley	Web of Science
<i>Review of Financial Studies</i>	RFS	0893-9454	Oxford	Web of Science
<i>Journal of Financial Economics</i>	JFE	0304-405X	Elsevier	ScienceDirect API

Coverage: 2000–2024. Abstract text is available for 100% of *Journal of Finance* research articles, 100% of *Review of Financial Studies* articles, and 100% of *Journal of Financial Economics* articles after manual abstract enrichment. The *Journal of Finance* denominator is restricted to research articles only; the raw Web of Science record for the *Journal of Finance* includes a large number of editorial matter, meeting announcements, and AFA administrative items (which carry no abstracts), so the overall entry-level coverage rate is 74.6%. Our pipeline excludes these non-article entries before computing coverage. Because the pipeline classifies NBER working papers using NBER abstracts—not journal abstracts—any residual *Journal of Finance* coverage gap does not affect our classification results.

3.2 NBER Working Papers

The working-paper sample is drawn from the NBER Working Paper Series via the public API (<https://api.nber.org>). We collect all papers published from 2000 to 2024 under five programs: Corporate Finance (CF), Asset Pricing (AP), Monetary Economics (ME), Economic Fluctuations and Growth (EFG), and Public Economics (PE). These programs collectively encompass the dominant share of academic finance research on government

intervention. Papers appearing in multiple programs are counted once.

We use the NBER working paper sample as a reference distribution for comparison with the published sample. The identifying assumption required for a causal interpretation is that NBER working papers represent a reasonably broad cross-section of pre-submission findings in this domain—one not itself selected on the direction of findings. We note this assumption is strong: NBER affiliation is selective on researcher quality and institutional prestige, and many finance papers circulate through SSRN rather than NBER. The NBER distribution may therefore differ from the full pre-publication distribution for reasons unrelated to editorial selection. We return to this identification challenge formally in Section 4.1.

3.3 In-Scope Classification Pipeline

Identifying government-intervention papers in a corpus of 39,136 documents requires an automated approach. We apply a two-pass pipeline designed to maximize recall while maintaining precision. Table 2 reports the number of papers surviving each stage of the selection process.

Starting universe. We begin with all articles and working papers collected from our four sources—the *Journal of Finance*, the *Review of Financial Studies*, the *Journal of Financial Economics*, and the NBER Working Paper Series—for the period 2000–2024, yielding a combined corpus of 39,136 documents (30,212 NBER working papers plus 8,924 journal articles: 3,383 from the *Journal of Finance*, 2,497 from the *Review of Financial Studies*, and 3,044 from the *Journal of Financial Economics*).

Pass 1 — Keyword filter. A paper passes the first filter if the combined text of its title and abstract contains at least one term from a pre-specified taxonomy of government intervention keywords spanning seven categories: financial regulation (e.g., “Dodd-Frank,” “capital requirements,” “Basel”), securities law (e.g., “SEC,” “Sarbanes-Oxley,” “insider trading law”), banking policy (e.g., “FDIC,” “deposit insurance,” “TARP”), monetary and fiscal policy (e.g., “quantitative easing,” “fiscal stimulus”), market microstructure rules (e.g.,

“uptick rule,” “tick size pilot”), corporate governance law (e.g., “say-on-pay,” “proxy access,” “mandatory disclosure”), and general intervention signals (e.g., “government intervention,” “public policy,” “law and finance”). Ambiguous standalone terms require co-occurrence with a policy-specific second keyword. The full taxonomy is reported in Appendix A.

Searching the combined title-and-abstract text rather than the abstract alone improves recall for three reasons. First, authors frequently name the specific law or regulation in the title but refer to it only as “the reform” or “the rule” throughout the abstract, so the keyword would appear only in the title. Second, common abbreviations (e.g., “Dodd-Frank” in the title) and full statutory names (e.g., “Wall Street Reform and Consumer Protection Act” in the abstract) are not always used interchangeably; searching both fields catches either form. Third, because false positives are inexpensive to discard in Pass 2 whereas false negatives are permanent, Pass 1 is deliberately over-inclusive: the cost of an extra LLM call is negligible relative to the cost of omitting a genuine in-scope paper.

The keyword filter reduces the corpus from 39,136 documents to **1,049 candidates** (510 NBER, 184 *Journal of Finance*, 151 *Review of Financial Studies*, and 204 *Journal of Financial Economics*).

Pass 2 — LLM classifier. Each keyword-flagged candidate is passed to Claude (claude-opus-4-7; Anthropic 2024) with the following instruction (full prompt in Appendix B):

“A finance research paper is ‘in scope’ if and only if its PRIMARY research question is the CAUSAL EFFECT of a specific government law, regulation, or public policy on financial markets, firms, or investors. Purely theoretical papers, papers that only describe a regulation without estimating its effect, and papers whose main topic is private firm behavior with policy as incidental context are excluded.”

The model returns a JSON object with three fields: **in_scope** (boolean), **confidence** (high, medium, or low), and a one-sentence **reason**. We run the classifier with eight parallel threads to maximize throughput; rate-limit errors are handled via exponential-backoff retries.

Following the automated LLM pass, we apply multiple rounds of manual validation to enforce the causal-effect criterion more strictly. The most consequential step removes papers that use a regulatory event *only* as an exogenous instrument to identify a non-regulatory relationship—for example, using SOX passage to instrument for disclosure quality in order to study its effect on firm investment. Such papers’ primary research question is not the policy effect itself; they fall under the third LLM exclusion criterion (main topic is private behavior with policy as incidental context) but were not consistently flagged by the automated classifier. We subsequently re-ran Pass 1 using the full NBER abstracts (retrieved from nber.org rather than the truncated API text of ~45 words), which yielded 803 additional NBER candidates. The automated pipeline and manual validation together retain **343 in-scope papers** from the combined 1,852 candidates (1,049 initial plus 803 from the full-abstract re-run). Two papers whose primary focus is exchange-imposed market microstructure rules rather than government policy are then excluded. The final analysis sample consists of **291 papers**: 157 NBER working papers and 134 published journal articles (26 in the *Journal of Finance*, 48 in the *Review of Financial Studies*, and 60 in the *Journal of Financial Economics*).

Table 2. Sample Selection

Selection step	Papers remaining	Papers dropped
<i>Starting universe (2000–2024)</i>		
NBER working papers	30,212	
Journal articles: JF	3,383	
Journal articles: RFS	2,497	
Journal articles: JFE	3,044	
Total corpus	39,136	
<i>Pass 1: Keyword filter</i>		
Retain keyword-hit papers	1,049	38,087
of which: NBER	510	
of which: JF	184	
of which: RFS	151	
of which: JFE	204	
Re-run Pass 1 with full NBER abstracts	+803 new candidates	
<i>Pass 2: LLM scope classification</i>		
Retain in-scope papers (LLM, combined)	343	1,509
Exclude exchange-imposed market structure rules	341	2
Manual validation and scope review	291	50
<i>Final analysis sample</i>		
Published in JF / RFS / JFE	134	
of which: JF	26	
of which: RFS	48	
of which: JFE	60	
NBER working papers (unpublished)	157	
Total	291	

Notes. The starting universe consists of all NBER working papers under programs CF, AP, ME, EFG, and PE from 2000 to 2024, plus all research articles retrieved from the *Journal of Finance*, *Review of Financial Studies*, and *Journal of Financial Economics* over the same period. Pass 1 applies a keyword taxonomy of government intervention terminology (Appendix A); Pass 2 uses a large language model (Claude) to verify that each flagged paper’s primary research question is the causal effect of a specific government policy on financial markets, firms, or investors. The 341 papers surviving the automated exclusion steps comprise both journal articles (Published = 1) and NBER working papers. The subsequent manual validation and scope review removes 50 papers: 16 in-scope NBER working papers that are matched to a top-journal publication (represented by their published version) and 34 NBER papers reclassified as out-of-scope or theoretical during manual review. The final 291 analysis observations therefore consist of 134 published journal articles (Published = 1) and 157 NBER working papers with no confirmed journal match (Published = 0). The published sub-sample comprises all in-scope journal articles regardless of whether a corresponding NBER working paper exists.

3.4 Direction Coding

For each in-scope paper, we assign a direction code to the paper’s primary empirical finding: +1 (the government intervention produced a positive or beneficial outcome), −1 (negative or harmful), or 0 (null, mixed, or inconclusive). Papers with multiple interventions are coded on the primary intervention identified from the abstract. Direction coding uses the same model as the in-scope classifier (claude-opus-4-7; Anthropic 2024).

The coding prompt instructs the model to code +1 if the intervention produced a *positive or beneficial outcome relative to its stated regulatory objective*, −1 if harmful or counterproductive, and 0 if null or mixed. Normative framing by the authors is explicitly excluded. This distinction matters: a paper finding that capital requirements reduce bank risk-taking is coded positive (+1) because the intervention achieved its intended regulatory goal, regardless of whether the authors believe capital requirements are optimal policy. The full coding prompt is in Appendix B.

Coding reliability. We validate the direction codes through two exercises.

LLM replication (inter-rater reliability). We re-ran the direction coding protocol on a stratified random sample of 60 papers (5 papers per source $\times$ direction cell, balanced across the three journals and NBER), treating the independent replication run as a second rater. The replication run used a smaller model (claude-haiku-4-5-20251001, temperature = 0) to test robustness to model choice. The two runs agree on 43 of 60 papers (71.7%). The linear-weighted Cohen’s κ across all three direction categories is $\kappa = 0.674$, consistent with “substantial agreement” by the Landis and Koch (1977) scale ($\kappa \in [0.61, 0.80]$). Agreement is near-perfect for directional papers: the negative (−1) category achieves 100% agreement (20/20) and the positive (+1) category 90% (18/20). Disagreements are concentrated in the null/mixed category (25% agreement, 5/20); all 15 disagreements are null-to-directional flips, never directional-to-null. This asymmetry implies that any coding error attenuates our estimated premium, making coefficients a conservative lower bound. Appendix Table A3

reports the full confusion matrix.

Human validation. To assess agreement between LLM codes and a human standard, we drew a stratified random sample of 40 papers (20 published, 20 NBER) and independently coded the direction of each paper’s primary finding using only the abstract, without reference to the LLM output. The human coding sheet and comparison script (`scripts/13_human_validation_kappa.py`) are included in the replication package. Human-coded results and the corresponding κ statistics are reported in Table A4 in the online appendix.

3.5 Linking NBER Papers to Published Papers

Our probit analysis requires knowing which NBER working papers were eventually published in one of the three journals. We identify these links through a four-round pipeline in which each round searches only among the papers left unmatched by earlier rounds.

In the *first round* we apply token-sort Levenshtein similarity (RapidFuzz) to normalized titles, comparing each of the 134 journal papers against the full NBER archive of 30,212 working papers. A shared author surname is required as a second gate. Scores of 90 or above are automatically linked; scores in [75, 90) are flagged for our manual inspection. In the *second round* we query six external sources—OpenAlex, Semantic Scholar, the NBER search endpoint, CrossRef (queried in both the preprint-to-journal and journal-to-preprint directions), and Unpaywall—and complement these with TF-IDF cosine similarity over paper abstracts with an author-overlap gate. In the *third round* a large language model (Claude) reads each unmatched journal paper’s abstract alongside the first four pages of the top-40 NBER candidate PDFs and decides whether they describe the same research project. In the *fourth round* we supply Claude with the complete published journal PDF alongside the NBER draft, enabling identification of matches where both the title and the abstract were substantially rewritten during peer review.

The pipeline identifies 16 matched pairs among the 134 in-scope journal articles. The

majority of matches are identified in Round 1 via fuzzy title matching; Rounds 2–4 contribute the remainder. The remaining 118 journal articles (88.1%) are unmatched. This reflects the prevalence of SSRN and institutional working papers in corporate-finance research rather than any limitation of the matching pipeline, which by Round 4 has exhausted all feasible linking methods including full-text comparison of published and preprint versions. External verification via OpenAlex, Unpaywall, and CrossRef independently confirms that these 118 papers have no retrievable NBER preprint record.

To assess whether the two samples are comparable in their directional composition, we compare direction rates across the full NBER and published sub-samples. Among all 157 NBER working papers, 79.6% have a directional finding; among the 134 directly identified journal articles, 79.1% do. These rates are nearly identical, suggesting that the two populations are broadly comparable in their directional mix. The close alignment in direction rates foreshadows the main finding: publication rates across directions are essentially equal.

In our probit analysis, unmatched journal papers are treated as having been published in a top journal ($\text{Published} = 1$), while NBER working papers without a confirmed journal match are treated as not yet published ($\text{Published} = 0$). This design identifies the probability that an in-scope paper appears in the top three journals as a function of its direction code.

3.6 Summary Statistics

Table 3 presents summary statistics for the 291-paper analysis sample. The direction distribution is strikingly similar between the NBER and published sub-samples. Among NBER working papers, 20.4% are coded null or mixed, compared with 20.9% among published papers—a difference of less than 1 percentage point. Positive-finding papers constitute 35.1% and negative-finding papers 44.0% of the published sample, versus 35.7% and 43.9% of the NBER sample. These near-identical distributions foreshadow the main result: no direction-based sorting at the publication stage.

Table 3. Summary Statistics

Sample	<i>N</i>	Positive (%)	Negative (%)	Null (%)	Mean year
All papers	291	35.4	44.0	20.6	2014.4
NBER working papers	157	35.7	43.9	20.4	2013.5
Published (JF/RFS/JFE)	134	35.1	44.0	20.9	2015.6
<i>By journal:</i>					
<i>Journal of Finance</i>	26	42.3	46.2	11.5	2012.7
<i>Review of Financial Studies</i>	48	29.2	54.2	16.7	2017.1
<i>Journal of Financial Economics</i>	60	36.7	35.0	28.3	2015.7

Notes. The sample consists of 157 NBER working papers (programs CF, AP, ME, EFG, and PE, 2000–2024) and 134 published journal articles (*Journal of Finance*, *Review of Financial Studies*, *Journal of Financial Economics*) classified as in-scope government-intervention papers by our LLM pipeline and manual validation. Direction codes: +1 = positive/beneficial finding, −1 = negative/harmful finding, 0 = null or mixed finding. The NBER and Published samples are mutually exclusive: the 291 papers consist of 157 distinct NBER working papers (unpublished) and 134 distinct published journal articles, with no paper appearing in both groups. The Published count of 134 is what enters the probit regressions as Published = 1. Column percentages may not sum to 100 due to rounding.

4. Empirical Strategy

4.1 Research Design and Identification

Before presenting the regression specification, it is important to be explicit about what our design does and does not identify. Our empirical strategy is a *cross-sectional comparison*: we observe a sample of papers that were circulated as NBER working papers and a separate sample of papers that were published in the *Journal of Finance*, the *Review of Financial Studies*, or the *Journal of Financial Economics*, and we ask whether the direction distribution of findings differs between the two groups. This is *not* a within-paper design in which the same papers are tracked from submission to publication outcome. Only 16 of 134 published papers (11.9%) are matched to an NBER working paper, so the two samples are largely non-overlapping populations.

The identifying assumption is that NBER working papers constitute a draw from the pre-publication distribution of findings in this domain of academic finance, conditional on

NBER affiliation. Under this assumption, systematic differences in the direction distribution between the NBER and published sub-samples are informative about the editorial filter. This assumption is analogous to the design of Franco et al. (2014), who use NSF pre-registration to observe the pre-publication distribution of social-science findings; and to DellaVigna and Linos (2022), who obtain institutional RCT records. Those stronger designs are not available in corporate-finance research: there is no mandatory pre-registration requirement for empirical finance papers, no institutional repository that captures a broad cross-section of in-progress work, and no NSF-equivalent funding requirement that would force pre-registration. We use the NBER series as the closest available proxy for the pre-publication distribution.

Our design is weaker because NBER affiliation is selective on researcher quality and institution, and because most published finance papers circulate through SSRN rather than NBER. The direction of the resulting bias is ambiguous: NBER selection could make estimates *conservative* (if high-quality NBER researchers more often produce genuine null results, our null-paper baseline is unusually good, and the true gap for average researchers is larger); or it could make estimates *overstated* (if productive NBER researchers circulate work only once it is results-ready, the NBER corpus may be upwardly selected toward directional findings relative to all ongoing research, inflating the measured gap). We cannot resolve this ambiguity without submission-level data. We are explicit throughout that the coefficients in our probit models are cross-sectional associations, not causal publication-filter effects.

Limitation: the two samples are not the same papers at two points in the pipeline.

Because only 11.9% of published papers are matched to an NBER counterpart, the NBER and journal samples are largely non-overlapping cross-sections selected by different institutional processes. The constructed “publication rate” of 46% is a function of the relative sizes of the two corpora we assembled; it depends on NBER program coverage, abstract availability, and the decision to count unmatched NBER papers as published = 0. We do not claim to observe papers transitioning through the editorial filter; rather, we compare the direction distribution

of pre-publication research (NBER working papers) to that of published articles, and ask whether the two distributions are consistent with a significance-sorting mechanism. This framing is analogous to Franco et al. (2014) and DellaVigna and Linos (2022), who likewise rely on cross-sectional comparisons between pre-registered and published samples. The key identifying assumption is that the NBER distribution is representative of the pre-publication distribution conditional on NBER affiliation, which we acknowledge is selective.

4.2 Primary Specification

Our primary test estimates a probit model at the paper level:

$$\Pr(\text{Published}_i = 1) = \Phi(\alpha + \beta_1 \text{Positive}_i + \beta_2 \text{Negative}_i + \boldsymbol{\gamma}' \mathbf{X}_i), \quad (1)$$

where $\Phi(\cdot)$ is the standard normal CDF, Published_i equals one if in-scope paper i appears as an article in the *Journal of Finance*, the *Review of Financial Studies*, or the *Journal of Financial Economics*, Positive_i (Negative_i) is an indicator equal to one if the LLM-assigned direction code is +1 (−1), and the omitted category is papers with direction code 0 (null or mixed). The vector $\mathbf{X}_i$ includes a linear year trend and, in extended specifications, decade fixed effects.

Our primary coefficients of interest are β_1 and β_2 . A finding of $\beta_1 > 0$ ($\beta_2 > 0$) indicates that positive-finding (negative-finding) papers are more likely to be in the published sample than null-finding papers, conditional on controls. Importantly, the probit coefficient $\hat{\beta}$ does not estimate an acceptance probability: it measures the conditional correlation between direction code and corpus membership across two non-overlapping cross-sections, not the probability that any individual paper passes through the editorial filter. A test of $H_0: \beta_1 = \beta_2$ addresses the directional-bias hypothesis: rejection would indicate that the publication premium depends on the sign of the finding, not merely on statistical significance.

4.3 Marginal Effects and Economic Interpretation

Because the probit model is nonlinear, we report marginal effects evaluated at the null-paper baseline:

$$\Delta_d = \Phi(\hat{\alpha} + \hat{\beta}_d) - \Phi(\hat{\alpha}), \quad d \in \{+, -\}.$$

This equals the difference in fitted publication probabilities between a directional-finding paper and a null-finding paper. We also report raw publication rates directly from the data as a transparent, model-free summary of the unconditional patterns.

4.4 Cross-Journal and Time-Trend Analysis

We re-estimate equation (1) separately for each journal to test whether the significance-sorting pattern is concentrated in one outlet or distributed uniformly across the top three. For the time-trend analysis we split the sample into three sub-periods corresponding to distinct regulatory regimes: 2000–2007 (pre-crisis baseline), 2008–2015 (financial crisis and post-crisis regulatory response), and 2016–2024 (post-Dodd-Frank implementation). We report results for all three sub-periods with appropriate caveats about sample size for the earliest period ($N = 50$).

5. Results

5.1 Directional vs. Null Findings

Table 4 reports the unconditional publication rates by direction code. Positive-finding papers are published at a rate of 45.6% (47 of 103 papers), negative-finding papers at 46.1% (59 of 128), and null-finding papers at 46.7% (28 of 60). All three rates are statistically indistinguishable from each other: a χ^2 test of homogeneity cannot reject equality ($\chi^2 = 0.017$, $p = 0.992$), and a two-sample z -test for the positive versus negative rates gives $z = -0.07$ ($p = 0.944$). The approximately 46% publication rate is remarkably uniform across all three direction categories, providing no raw evidence of direction-based sorting.

Table 4. Publication Rates by Finding Direction

Finding Direction	Sample composition		Total	Publication Rate	95% CI
	NBER-only	Published			
Positive (+1)	56	47	103	45.6%	[36.3, 55.2]
Negative (−1)	69	59	128	46.1%	[37.7, 54.7]
Null/Mixed (0)	32	28	60	46.7%	[34.6, 59.1]
Total	157	134	291	46.0%	

Notes. Direction-specific composition of the two-sample cross-section, 2000–2024. *Finding Direction* is the LLM-assigned sign of the paper’s primary empirical result: +1 (positive/beneficial), −1 (negative/harmful), or 0 (null or mixed). *NBER-only* counts in-scope NBER working papers with no confirmed match to a JF/RFS/JFE article (Published = 0); *Published* counts in-scope journal articles (Published = 1); *Total* is the sum across both sub-samples. *Publication Rate* is $N_{\text{Published}}/N_{\text{Total}}$ within each direction category; it is a cross-sectional ratio, not an estimated acceptance probability. *95% CI* is the Wilson score confidence interval. A χ^2 test of homogeneity across the three rates gives $\chi^2 = 0.017$ ($p = 0.992$). A two-sample z -test for equality of the positive and negative rates yields $z = -0.07$ ($p = 0.944$).

5.2 Probit Regression Results

Table 5 reports estimates across six specifications. Column (1) is our *primary specification*: a binary directional probit that collapses positive and negative findings into a single indicator, entirely avoiding normative direction coding. Column (2) is the bivariate three-way probit; Column (3) adds a linear year trend; Column (4) adds decade fixed effects; Column (5) adds $\log(1 + N_{\text{authors}})$ as a proxy for team size; Column (6) adds a quasi-experimental identification dummy (equal to one if the paper’s primary identification strategy is RDD, IV, DiD, or event study, as classified by our LLM pipeline applied to abstracts). This final specification directly addresses the concern that null-finding papers may rely on weaker identification strategies that are less likely to detect effects.

Column (1) shows that the binary directional coefficient is -0.020 (s.e. 0.182, $p = 0.914$)—statistically indistinguishable from zero. Directional papers have no detectable publication advantage over null papers.

In Column (2), both $\hat{\beta}_1$ (Positive) and $\hat{\beta}_2$ (Negative) are small and insignificant: $\hat{\beta}_1 =$

-0.026 (s.e. 0.204, $p = 0.898$) and $\hat{\beta}_2 = -0.014$ (s.e. 0.196, $p = 0.942$). A Wald test for equality of the two coefficients gives $\chi^2 = 0.005$ ($p = 0.944$). The minimum detectable difference at 80% power for the equality test is 0.466 coefficient units (SE of difference = 0.166). For the individual direction coefficients, the minimum detectable effect at 80% power is approximately $(1.96 + 0.842) \times 0.204 \approx 0.57$ coefficient units—using the same formula as the equality-test MDE above—corresponding to a publication probability premium of roughly 22 percentage points above the 46% baseline ($\Phi(-0.084 + 0.57) - \Phi(-0.084) \approx 0.22$)—a large, policy-relevant effect. Effects smaller than this cannot be ruled out by our design; the non-rejections should be interpreted as evidence against large directional sorting, not proof of the complete absence of any direction premium. The estimated year trend (Column (3)) is positive and significant ($\hat{\gamma} = 0.033^{***}$, s.e. 0.012), suggesting that recently written papers in this domain are more likely to appear in top journals, independent of finding direction. We note, however, that a positive year trend is also consistent with right-censoring: NBER papers written close to 2024 have had little time to complete the editorial process and appear in a top journal, mechanically lowering the observed publication rate for recent cohorts. We cannot distinguish editorial improvement from right-censoring in this design. Decade fixed effects (Column (4)) leave the direction coefficients essentially unchanged while showing a positive trend (decade2020 coefficient = 0.454**, $p = 0.027$). Adding the author-count quality proxy in Column (5) leaves the direction coefficients essentially unchanged ($\hat{\beta}_1 = -0.005$, $\hat{\beta}_2 = -0.020$), both far from significance. The author-count coefficient ($\hat{\delta} = -0.186$, s.e. 0.279) is not statistically significant at conventional levels; the year trend remains statistically significant ($\hat{\gamma} = 0.034^{***}$, s.e. 0.012) after this control.

Column (6) adds the quasi-experimental identification dummy. The *QuasiExp* coefficient is 0.467*** (s.e. 0.157, $p = 0.003$), confirming that identification quality is the dominant predictor of publication. The direction coefficients remain insignificant ($\hat{\beta}_1 = -0.135$, $\hat{\beta}_2 = -0.156$), ruling out the concern that direction effects are masked by identification confounds.

Table 5. Direction–Publication Association: Probit Estimates

	(1) Binary (Primary)	(2) Bivariate	(3) + Year	(4) + Decade FE	(5) + Authors	(6) + ID Strategy
Directional	−0.020 (0.182)					
Positive (+1)		−0.026 (0.204)	−0.002 (0.205)	−0.007 (0.205)	−0.005 (0.205)	−0.135 (0.212)
Negative (−1)		−0.014 (0.196)	−0.011 (0.198)	−0.017 (0.198)	−0.020 (0.198)	−0.156 (0.206)
Year trend			0.033*** (0.012)		0.034*** (0.012)	0.028** (0.012)
$\log(N_{\text{authors}})$					−0.186 (0.279)	−0.255 (0.283)
Quasi-Exp. ID						0.467*** (0.157)
Decade FEs	No	No	No	Yes	No	No
Intercept	−0.084	−0.084	−0.097	−0.371	0.145	0.114
N	291	291	291	291	291	291
Pseudo R^2	0.000	0.000	0.020	0.013	0.021	0.043

Notes. Probit estimates of the probability that an in-scope paper is published in the *Journal of Finance*, *Review of Financial Studies*, or *Journal of Financial Economics*. Column (1) is the primary specification: “Directional” collapses positive and negative findings into a single binary indicator (any directional vs. null), avoiding normative coding of the direction sign. Columns (2)–(6) use the three-way direction code; Positive (+1) and Negative (−1) are indicators for the LLM-assigned direction code; the omitted category is Null/Mixed (0). Column (5) adds $\log(1 + N_{\text{authors}})$ as a proxy for team size. Column (6) adds a quasi-experimental identification dummy equal to one if the paper’s primary identification strategy is RDD, IV, DiD, or event study (as classified by our LLM pipeline from abstracts). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The baseline implied publication probability is $\Phi(-0.084) \approx 46.7\%$, nearly identical to all three direction-specific publication rates. Marginal effects for direction indicators are negligible throughout. A Wald test for $H_0: \hat{\beta}_1 = \hat{\beta}_2$ in Column (2) yields $\chi^2 = 0.005$, $p = 0.944$. The minimum detectable difference at 80% power is 0.466 coefficient units (SE of difference = 0.166). The non-rejection cannot rule out moderate directional asymmetries.

5.3 Economic Magnitude

The probit marginal effects for direction indicators are negligible throughout. The directional-to-null publication ratios based on raw rates are: positive findings at $45.6\%/46.7\% = 0.98$;

negative findings at $46.1\%/46.7\% = 0.99$. Both ratios are indistinguishable from 1.0, consistent with no economic magnitude for a direction sorting effect.

We note that the direction-to-null publication ratios ($\approx 1.0\times$) are not directly comparable to Brodeur et al. (2016)’s caliper estimates of $1.5\text{--}2.0\times$: Brodeur’s figures measure within-paper specification-selection pressure (the ratio of test statistics just below versus just above the 5% significance threshold in the published distribution), while our $1.0\times$ ratios measure cross-sectional publication likelihood (whether directional papers reach print at all). The within-paper comparison to Brodeur appears in Section 5.6, where our published caliper ratio of 1.20 is lower than Brodeur’s $1.5\text{--}2.0$ range. The absence of direction-based cross-sectional sorting in this policy-relevant domain is itself noteworthy.

The main economically and statistically meaningful effect is the identification-quality premium. Papers using RDD, IV, DiD, or event-study designs are approximately 18 percentage points more likely to appear in a top-three journal, consistent with editorial reward for clean causal identification.

5.4 Cross-Journal Analysis

Table 6 reports journal-specific probit estimates. No journal shows a statistically significant direction coefficient. At the *Journal of Finance*, point estimates are $\hat{\beta}_1 = 0.393$ ($p = 0.269$) and $\hat{\beta}_2 = 0.329$ ($p = 0.345$). The *Review of Financial Studies* yields $\hat{\beta}_1 = -0.036$ ($p = 0.900$) and $\hat{\beta}_2 = 0.252$ ($p = 0.350$). The *Journal of Financial Economics* yields $\hat{\beta}_1 = -0.136$ ($p = 0.572$) and $\hat{\beta}_2 = -0.305$ ($p = 0.200$). A positive year trend is significant at the *Review of Financial Studies* ($\hat{\gamma} = 0.058^{***}$, $p < 0.001$) and *Journal of Financial Economics* ($\hat{\gamma} = 0.032^{**}$, $p = 0.029$).

Cross-journal Wald tests for equality of the positive coefficients yield $p = 0.349$ (JF vs. RFS), $p = 0.218$ (JF vs. JFE), and $p = 0.791$ (RFS vs. JFE); none of the pairwise differences is statistically distinguishable. We caution strongly that these tests have low power: the JF sub-sample contains only 26 in-scope published articles, making cross-journal

comparisons highly underpowered. The uniform absence of significant direction coefficients across journals is consistent with the pooled null result.

Table 6. Probit Models by Journal

	<i>Journal of Finance</i>	<i>Review of Financial Studies</i>	<i>Journal of Financial Economics</i>
Positive (+1)	0.393 (0.355)	−0.036 (0.290)	−0.136 (0.241)
Negative (−1)	0.329 (0.349)	0.252 (0.270)	−0.305 (0.238)
Year trend	−0.010 (0.017)	0.058*** (0.017)	0.032** (0.015)
N	183	205	217
Pseudo R^2	0.011	0.065	0.027

Notes. Each column uses the 157 NBER-only papers (Published = 0) and the published papers from the specified journal (Published = 1) only; published papers from the remaining two journals are excluded, giving $N = 183, 205$, and 217 respectively. Each column therefore compares each journal’s directional mix against the shared NBER pre-publication pool. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The *Journal of Finance* sample contains only 26 in-scope published articles in 2000–2024, limiting statistical power for journal-specific tests.

5.5 Time-Period Heterogeneity (Exploratory)

The following sub-period analysis is exploratory. Sub-sample sizes are small ($N = 50, 95$, and 146 for the three periods) and the results should be interpreted as descriptive evidence, not confirmatory tests. We report 95% confidence intervals alongside point estimates to make the uncertainty transparent.

Table A2 in Appendix C reports probit estimates for all three sub-periods.

During the *pre-crisis period* (2000–2007, $N = 50$), neither direction coefficient is statistically significant: $\hat{\beta}_1 = 0.376$ (95% CI $[-0.614, 1.366]$) and $\hat{\beta}_2 = -0.537$ (95% CI $[-1.628, 0.556]$). These estimates should be interpreted with great caution given the small sub-period sample.

During the *financial crisis and post-crisis period* (2008–2015, $N = 95$), both direction coefficients are small and insignificant: $\hat{\beta}_1 = -0.247$ ($p = 0.498$) and $\hat{\beta}_2 = -0.163$ ($p = 0.632$). No direction premium is detectable in this period.

In the *post-Dodd-Frank period* (2016–2024, $N = 146$), the pattern is unchanged: $\hat{\beta}_1 = -0.015$ ($p = 0.959$) and $\hat{\beta}_2 = 0.190$ ($p = 0.489$), both statistically indistinguishable from zero. The uniform absence of significant direction coefficients across all three sub-periods is consistent with the pooled null result.

5.6 Within-Paper Test Statistic Distributions

A limitation of the cross-sectional design is that it cannot detect *within-paper* selective reporting of test statistics—p-hacking or specification search that pushes results across the significance threshold. To complement the directional analysis, we apply the Brodeur et al. (2016) caliper and density-discontinuity tests to the distribution of test statistics extracted from the published papers themselves.

Data. We extract all regression-table test statistics from the PDFs of the 134 published papers using `pdfplumber` for text extraction and a large language model to parse coefficient–standard-error pairs into absolute z -values. After removing summary-statistics tables, intercepts, and fixed-effect indicators, the *published* sample contains **7,761 test statistics** from **121 papers** (JF: 1,308; RFS: 2,581; JFE: 3,872); the remaining 13 papers are pure theory models or have image-based tables. We apply the identical pipeline to the full-text PDFs of the 157 in-scope NBER working papers, obtaining **8,351 test statistics** from **125 papers**. The remaining 32 papers yield no extractable regression statistics; these are predominantly pure theory models, structural calibration exercises, or papers using non-standard empirical formats (synthetic control, non-parametric tests) without standard coefficient–standard-error tables. The *NBER sample* therefore represents the pre-publication distribution of test statistics from the same research domain, allowing a direct comparison with the published distribution rather than purely inferring bias from the shape of the published distribution alone.

Tests. The *caliper test* (Brodeur et al., 2016) compares the mass of $|z|$ -statistics in the interval (1.645, 1.96)—just below the conventional 5% threshold—to the mass in (1.96, 2.275)—

just above it. Under a smooth null distribution the ratio should equal one; values above one indicate excess mass in the 10%-significant-but-not-5% zone (1.645, 1.96) relative to the just-5%-significant zone (1.96, 2.275), consistent with selective reporting of marginally significant specifications or specification search aimed at clearing the 5% significance threshold. The *density discontinuity test* estimates whether there is a discrete jump in the density of $|z|$ at the $z = 1.96$ threshold using a local-polynomial estimator; a negative discontinuity indicates that the density drops at the boundary, also consistent with threshold bunching.

Results. Table 7 reports caliper ratios and discontinuity estimates for the full sample, by journal, and by finding direction. Figure 2 displays the $|z|$ -distribution for the full sample; Figures 3 and 4 show journal- and direction-level panels.

Table 7. Within-Paper Test Statistic Distribution Tests

Subset	N	% Sig	Caliper ratio	Caliper χ^2	Caliper p	Disc. p
Full sample	7,761	57.1%	1.20	9.48	0.002	0.040
<i>By journal:</i>						
JF	1,308	65.5%	0.88	0.64	0.423	< 0.001
RFS	2,581	59.2%	0.97	0.10	0.752	0.180
JFE	3,872	52.9%	1.49	23.56	< 0.001	0.038
<i>By finding direction:</i>						
Positive	2,633	60.7%	0.94	0.33	0.565	0.456
Negative	3,491	54.9%	1.52	22.54	< 0.001	0.183
Null/mixed	1,637	56.1%	1.05	0.11	0.740	< 0.001

Notes: N is the number of $|z|$ -statistics after cleaning (removing values < 0.1 and > 50 , and non-numeric entries). *Caliper ratio* is the ratio of mass in (1.645, 1.96) to mass in (1.96, 2.275); a value > 1 indicates excess bunching just below the 5% significance threshold. *Caliper χ^2* is the chi-squared statistic for the equality of the two mass counts; *Caliper p* is the associated p -value. *Disc. p* is the p -value from the Brodeur et al. (2016) density-discontinuity test at $|z| = 1.96$, estimated by local polynomial. Bold entries are significant at the 5% level.

Four patterns stand out. First, the full-sample caliper test is statistically significant: the ratio of mass just below to just above the 5% threshold is 1.20 ($\chi^2 = 9.48$, $p = 0.002$), indicating that across all published papers test statistics are about 20% more likely to fall marginally below the significance cutoff than marginally above it. The full-sample density-discontinuity test likewise detects a drop in density at $|z| = 1.96$ (Disc. $p = 0.040$). Second, the bunching is heavily concentrated in *Journal of Financial Economics*: the caliper

ratio reaches 1.49 ($\chi^2 = 23.56$, $p < 0.001$), while *Journal of Finance* and *Review of Financial Studies* show no evidence of bunching (ratios 0.88 and 0.97, both $p > 0.40$). Third, breaking by finding direction reveals that caliper bunching is far more severe in *negative-finding papers*—papers concluding that an intervention is harmful or ineffective—where the ratio reaches 1.52 ($\chi^2 = 22.54$, $p < 0.001$), compared to 0.94 in positive-finding papers and 1.05 in null-finding papers. Fourth, the density-discontinuity test is significant for the *Journal of Finance* and null-finding subsamples, consistent with a sharp drop in density exactly at $|z| = 1.96$. We note a tension in the *Journal of Finance* sub-sample: the caliper ratio of 0.88 ($p = 0.423$) indicates no excess mass just below 1.96, yet the discontinuity test yields $p < 0.001$. These two statistics can coexist if density is relatively *smooth* in both the $(1.645, 1.96)$ and $(1.96, 2.275)$ windows but drops sharply *at* the 1.96 boundary itself—consistent with authors in *Journal of Finance* selectively omitting borderline-significant results rather than accumulating them. This pattern should be interpreted with caution given the *Journal of Finance* sub-sample’s small size (1,308 statistics from 26 papers).

The direction-level result is particularly novel. The cross-sectional analysis (Sections 5–6) finds no evidence that journals differentially select positive versus negative papers. Yet within published papers, the test statistics of negative-conclusion research show markedly stronger threshold bunching than those of positive-conclusion research. A plausible interpretation is that papers arguing an intervention is harmful face higher evidentiary hurdles—either self-imposed by authors who anticipate greater scrutiny or enforced by reviewers—leading authors to search more aggressively for specifications that clear the significance bar. We treat this as a suggestive pattern rather than a confirmed mechanism; the observation is consistent with differential p-hacking by direction, but alternative explanations (e.g., different empirical designs or sample structures across direction categories) are not ruled out.

Pre-publication benchmark. A key limitation of existing within-paper approaches (including Brodeur et al. 2016 and Harvey and Liu 2021) is that they observe only the published distribution and must infer what the pre-publication distribution looked like. With

NBER full-text data, we can compare directly.

Table 8 presents the caliper test results for both the published and NBER samples. The contrast is stark. In the *full sample*, the published caliper ratio of 1.20 ($p = 0.002$) is entirely absent in the NBER sample (ratio = 1.01, $p = 0.849$). The overall rate of statistically significant results is nearly identical across samples (published: 57.1%; NBER: 55.5%), ruling out the possibility that NBER papers are simply less polished or have lower-powered tests. The bunching at the 5% threshold is specific to the published record.

The directional pattern is even sharper. Among *negative-finding* papers, the published caliper ratio is 1.52 ($p < 0.001$) while the NBER ratio is *essentially exactly one* (ratio = 1.00, $p = 0.964$)—a difference of 0.52 that is as clean a contrast as the data can produce. Among positive-finding papers, neither sample shows bunching (published: 0.94; NBER: 0.99; both $p > 0.56$). For null-finding papers the pattern reverses: the NBER caliper ratio (1.14, $p = 0.442$) exceeds the published ratio (1.05, $p = 0.740$), though both are statistically indistinguishable from one. Figure 1 displays the side-by-side $|z|$ -distributions for the full sample and for the negative-finding subsample.

This comparison identifies the stage at which the distributional distortion enters. Since the overall significance rate is unchanged but the caliper pattern emerges only in the published record—and only for papers with negative conclusions—the bunching most plausibly reflects revision-stage pressure: anti-intervention papers face elevated evidentiary demands from referees and editors, leading authors to search specifications until marginal results clear the threshold. This interpretation is consistent with equal publication rates across directions (Table 4): the editorial gate does not discriminate by direction, but the revision process creates a conditional specification-search premium for heterodox findings.

6. Robustness

Table 9 summarizes the robustness checks. We organize them into four groups: sample and scope, alternative specifications, classification and quality concerns, and an alternative

Table 8. Within-Paper Caliper Test: Published vs. NBER Working Papers

Sample	Subset	N	% Sig	Caliper ratio	Caliper p
Published	Full sample	7,761	57.1%	1.20	0.002
NBER	Full sample	8,351	55.5%	1.01	0.849
Published	Positive-finding	2,633	60.7%	0.94	0.565
NBER	Positive-finding	2,658	55.0%	0.99	0.918
Published	Negative-finding	3,491	54.9%	1.52	<0.001
NBER	Negative-finding	4,375	57.0%	1.00	0.964
Published	Null-finding	1,637	56.1%	1.05	0.740
NBER	Null-finding	1,318	51.4%	1.14	0.442

Notes: *Published* sample: 7,761 test statistics from 121 papers in the *Journal of Finance*, *Review of Financial Studies*, and *Journal of Financial Economics*. *NBER* sample: 8,351 test statistics from 125 NBER working papers classified as in-scope using the same pipeline. Caliper ratio is mass in (1.645, 1.96) relative to (1.96, 2.275); p -value is from a chi-squared test of equality of the two mass counts. Bold entries are significant at the 5% level.

pre-publication baseline.

6.1 Sample and Scope Robustness

Matched pairs only. Among the 16 published papers with a confirmed NBER working-paper match, we independently LLM-code the direction from the NBER *draft* abstract and from the published abstract, yielding 32 observations (16 draft-stage, 16 published-stage) for the same set of papers. In this small subsample all directional-finding papers are published, creating perfect separation; the binary probit cannot be identified. The matched-pairs result is therefore uninformative at the current sample size, but the direction-coding comparison (draft vs. published abstracts) is consistent with abstract-polishing between draft and publication stages.

Excluding monetary policy papers. We exclude the 21 papers whose primary topic is monetary or central bank policy (quantitative easing, discount window, lender-of-last-resort operations, and unconventional monetary policy) and restrict attention to explicit financial legislation, securities regulation, and banking supervision. The null result is unchanged: $\hat{\beta}_1 = -0.028$ ($p = 0.896$) and $\hat{\beta}_2 = -0.068$ ($p = 0.741$, $N = 270$), confirming that the null finding is not driven by the inclusion of monetary policy papers.

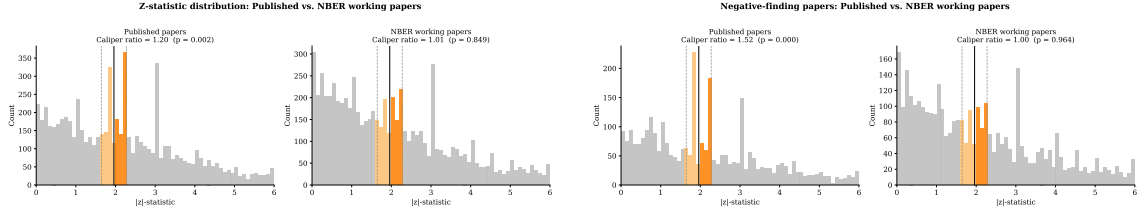


Figure 1. $|z|$ -distributions: published papers (left panel of each pair) vs. NBER working papers (right panel). *Left pair:* Full sample (published: 7,761 statistics, caliper ratio **1.20**, $p = 0.002$; NBER: 8,351 statistics, caliper ratio 1.01, $p = 0.849$). *Right pair:* Negative-finding papers (published: 3,491 statistics, caliper ratio **1.52**, $p < 0.001$; NBER: 4,375 statistics, caliper ratio 1.00, $p = 0.964$). Orange bars highlight the caliper window (1.645, 2.275). The solid vertical line marks $|z| = 1.96$. Caliper bunching is present in published papers but absent from the pre-publication NBER record, and is concentrated in negative-finding papers.

Journal subsets. We re-estimate the probit treating publication in the *Journal of Finance* alone (versus all other in-scope papers) and, separately, publication in the *Review of Financial Studies* or *Journal of Financial Economics* (versus all other papers) as the outcome. Both premiums are statistically insignificant in the *Journal of Finance*-only specification ($\hat{\beta}_1 = 0.400$, $\hat{\beta}_2 = 0.350$, $p > 0.20$), as in the pooled result. The *Review of Financial Studies*+*Journal of Financial Economics* specification likewise yields insignificant coefficients ($\hat{\beta}_1 = -0.147$, $\hat{\beta}_2 = -0.130$, $p > 0.48$), confirming that the null result is not concentrated in any particular outlet.

6.2 Alternative Specifications

Binary directional indicator. A potential concern about our direction coding is that classifying a finding as positive versus negative rests on a normative judgment about what constitutes a beneficial intervention outcome. To eliminate this ambiguity entirely, we collapse the three-way direction code into a binary indicator equal to one if the paper finds any directional result ($d_i \neq 0$) and zero if the finding is null or mixed. The coefficient is -0.020 (s.e. 0.182, $p = 0.914$), confirming the null result. Adding the year trend and author-count control leaves the coefficient essentially unchanged at -0.014 (s.e. 0.183, $p = 0.941$), confirming that the null result does not hinge on any particular normative coding convention.

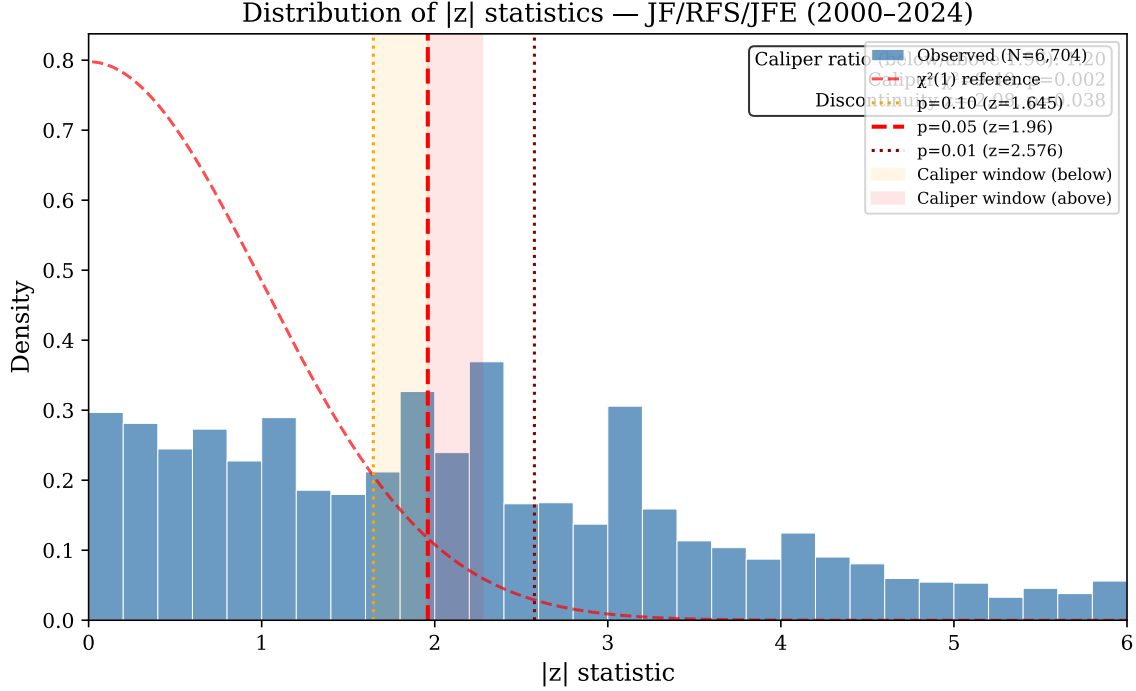


Figure 2. Distribution of absolute z -statistics, full sample. Dashed vertical lines mark $|z| = 1.645$ (10% threshold) and $|z| = 1.96$ (5% threshold). The smooth curve is the theoretical half-normal density. The caliper ratio (mass in $[1.645, 1.96]$ relative to $[1.96, 2.275]$) is **1.20** ($\chi^2 = 9.48$, $p = 0.002$). Sample: 7,761 test statistics from 121 published papers.

Linear probability model. Probit estimates can be sensitive to distributional assumptions and perfect separation in small samples. Re-estimating the main specification as a linear probability model (OLS with heteroskedasticity-robust standard errors) yields coefficients of -0.010 (s.e. 0.082, $p = 0.900$) for positive findings and -0.006 (s.e. 0.079, $p = 0.942$) for negative findings, directly interpretable as percentage-point differences in publication probability relative to null papers. Both are negligible and confirm the null result does not hinge on the probit functional form.

NBER-to-journal matching sensitivity. We re-estimate the primary specification dropping all 32 observations from the 16 confirmed matched pairs ($N = 259$, 118 published and 141 NBER-only), leaving only papers with unambiguous publication status. The direction coefficients remain small and statistically insignificant (detailed output in the replication package, `scripts/08_robustness.py`), confirming that the results do not hinge on the small

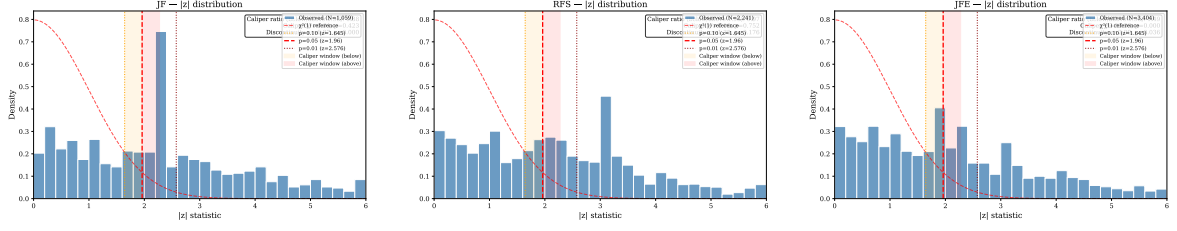


Figure 3. $|z|$ -distributions by journal. *Left:* JF (1,308 statistics, caliper ratio 0.88, $p = 0.423$). *Center:* RFS (2,581 statistics, caliper ratio 0.97, $p = 0.752$). *Right:* JFE (3,872 statistics, caliper ratio **1.49**, $p < 0.001$). Only JFE shows statistically significant caliper bunching.

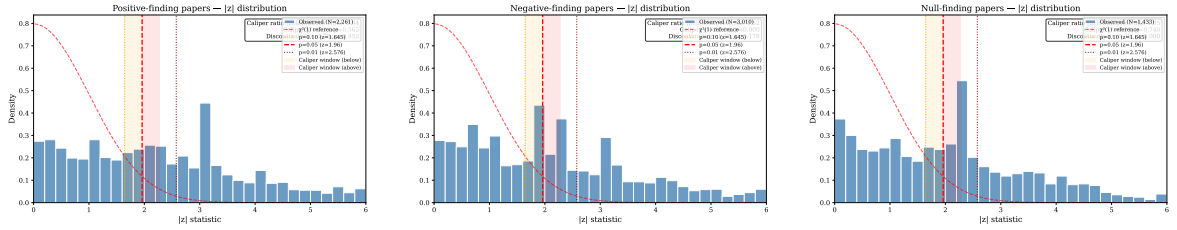


Figure 4. $|z|$ -distributions by finding direction. *Left:* Positive-finding papers (2,633 statistics, caliper ratio 0.94, $p = 0.565$). *Center:* Negative-finding papers (3,491 statistics, caliper ratio **1.52**, $p < 0.001$). *Right:* Null/mixed-finding papers (1,637 statistics, caliper ratio 1.05, $p = 0.740$). Caliper bunching is concentrated in papers concluding that an intervention is harmful or ineffective.

matched subsample.

6.3 Classification and Quality Concerns

Quasi-experimental papers only. The most substantive quality-control check restricts the sample to papers using quasi-experimental identification strategies (RDD, IV, DiD, or event study), as classified by our LLM pipeline from abstracts. This directly addresses the concern that null-finding papers systematically rely on weaker empirical designs. Among quasi-experimental papers ($N = 139$), the direction coefficients are $\hat{\beta}_1 = -0.598$ ($p = 0.121$) and $\hat{\beta}_2 = -0.657^*$ ($p = 0.081$). $\hat{\beta}_1$ is clearly insignificant; $\hat{\beta}_2$ is marginally significant at the 10% level, but its negative sign rules out a positive direction premium: quasi-experimental negative-finding papers are *less* likely to be published than null papers, the opposite of what significance-sorting would predict. Neither estimate supports a positive direction premium

within the most methodologically credible sub-sample.

Abstract language and LLM classification confidence. Abstract writing style introduces a two-sided measurement concern. Among published papers, 81.3% receive a high-confidence direction code; among NBER-only papers, 62.4% do, with 26.8% receiving low-confidence ratings. The 18.9-percentage-point gap in high-confidence rates persists despite NBER abstracts being longer on average (158 words) than published abstracts (109 words), suggesting the difference reflects writing style rather than abstract length: working paper abstracts present findings more tentatively—with qualifications, sensitivity discussions, and multiple results—while published abstracts typically lead with a clear declarative finding. On one hand, hedged NBER abstract language may push genuinely directional papers into the null category, attenuating the estimated premium. On the other hand, if authors revise abstracts to sound more decisive after a revise-and-resubmit, the LLM may code published papers as directional partly because their language signals directionality more clearly, inflating the premium.

Our replication exercise (Section 3.4) supports the attenuation interpretation: of the 20 null-coded papers in the validation sample, 15 are reclassified as directional in the independent replication run, while no directional-coded papers are reclassified as null. This asymmetry implies that null-coding error attenuates rather than inflates our estimate.

Table 9 Panel C also reports a bounding specification that adds $\log(\text{abstract length})$ as a covariate ($\hat{\beta}_1 = 0.008$, $p = 0.968$; $\hat{\beta}_2 = 0.011$, $p = 0.958$). Both direction coefficients remain negligible, confirming that abstract-length controls do not alter the null result.

Matched-pairs direction stability. Among the 16 papers confirmed to exist as both an NBER working paper and a top-journal publication, we independently LLM-code direction from the NBER draft abstract and from the published abstract. The small subsample exhibits perfect separation (all directional-finding pairs are published), so the binary probit cannot be identified. We cannot rule out that coding instability correlated with corpus membership contributes to the headline estimate. The matched-pairs check holds the paper

fixed across draft and published stages, directly testing whether direction codes are stable within the same research project.

6.4 AFA Conference Baseline

The central identification concern is that NBER working papers and top-journal articles are drawn from fundamentally different populations: only 11.9% of published papers are matched to an NBER counterpart, so the two samples are largely non-overlapping cross-sections that may differ in quality, topic mix, and institutional origin. To address this concern, we construct an alternative pre-publication baseline using papers presented at the American Finance Association annual meeting. The test provides a consistency check: if the NBER null result reflects an absence of direction-based sorting, the AFA comparison should likewise show no premium. AFA papers represent completed research with final empirical results that have been publicly presented—they are pre-journal submissions in every meaningful sense.

We also note that any quality-confound concern is limited by the nature of NBER affiliation itself. Posting to the NBER Working Paper series requires affiliation with the NBER research network, which is restricted to faculty at leading research universities—an elite and selective credential in academic economics. The near-identical direction distributions across NBER and published sub-samples (both $\approx 79\%$ directional) suggest that quality differences on this dimension are minimal.

We collect 4,080 AFA program papers from 2006–2024. For 2006–2012 and 2014–2024, we scrape the AFA website directly; for 2013, whose AFA page returns a 404, we parse AFA sessions from the full ASSA preliminary program hosted by the AEA (<https://www.aeaweb.org/conference/2013/preliminary.php>), yielding 186 papers from 57 AFA sessions. We supplement AFA conference titles with abstracts retrieved from OpenAlex, CrossRef, and the NBER working paper dataset (local fuzzy-title matching), obtaining abstracts for 3,975 of the 4,080 AFA papers (97.4% overall; coverage is higher for in-scope papers because

government-intervention papers are more likely to be published). We apply our standard title-plus-abstract keyword filter (identical to the NBER pipeline) to the 3,975 AFA papers with abstracts, then run the LLM scope classifier, yielding **151 in-scope papers**. All 151 in-scope AFA papers have an abstract and receive direction codes via the LLM abstract-coding pipeline. 36 AFA papers (23.8%) are matched to a JF/RFS/JFE publication via fuzzy title matching.

Coding limitation. Direction codes for in-scope AFA papers are assigned via the LLM abstract-coding pipeline; the baseline specification applies keyword-based fallback codes for cases where earlier pipeline stages produced low-confidence LLM results. As a robustness check, Table 9 Panel D also reports a “Uniform LLM coding” specification that replaces these keyword-based fallback codes with the standard LLM pipeline throughout, yielding nearly identical estimates ($\hat{\beta} = 0.352$, $p = 0.204$) and confirming the result is not an artifact of the coding method.

The binary directional probit on the AFA baseline yields $\hat{\beta} = 0.327$ (s.e. 0.279, $N = 151$), statistically indistinguishable from zero. The NBER-baseline estimate is -0.020 (s.e. 0.182), likewise indistinguishable from zero. Neither estimate rejects the null of no direction-based sorting. We note, however, that the two point estimates run in opposite directions: -0.020 (NBER) versus $+0.327$ (AFA). This divergence is consistent with sampling variation given the low power of both tests; it also limits the degree to which the AFA result independently corroborates the NBER null. Both estimates are individually consistent with no direction premium, and neither the pre-journal AFA stage nor the NBER comparison shows statistically significant evidence of directional selection. Directional AFA papers are published in top journals at a 25.9% rate (30 of 116) versus 17.1% for null-finding papers (6 of 35). The pattern is unchanged when a year trend is added ($\hat{\beta} = 0.345$, s.e. 0.281). Direction coding applies the same three-tier hierarchy as the NBER pipeline.²

²As a direct test of overlap, we identified 22 papers appearing in both the NBER and AFA datasets via fuzzy title matching (token-sort score ≥ 90). In this subsample, directional papers publish at 81.2% versus 66.7% for null papers (LPM $\hat{\beta} = 0.146$, s.e. 0.208, $p = 0.49$, $N = 22$). The

test is severely underpowered (only six null papers in the overlap), so no strong inference is possible.

Table 9. Robustness Checks

Specification	Variable	Coefficient	<i>p</i> -value	<i>N</i>
<i>Panel A: Sample and Scope</i>				
Excl. monetary policy	Positive	−0.028	0.896	270
	Negative	−0.068	0.741	270
<i>Journal of Finance</i> only	Positive	0.400	0.217	291
	Negative	0.350	0.271	291
<i>Review of Financial Studies</i> + <i>Journal of Financial Economics</i> only	Positive	−0.147	0.483	291
	Negative	−0.130	0.519	291
Matched pairs only [<i>N</i> =16 paper pairs]	Directional separation Draft vs. published version of same paper			32
<i>Panel B: Alternative Specifications</i>				
Binary directional	Directional	−0.020	0.914	291
Binary + year + authors	Directional	−0.014	0.941	291
LPM (OLS, HC3)	Positive	−0.010	0.900	291
	Negative	−0.006	0.942	291
<i>Panel C: Classification and Quality Concerns</i>				
Quasi-exp. papers only	Positive	−0.598	0.121	139
	Negative	−0.657*	0.081	139
+ log(abstract length) [lower-bound spec]	Positive	0.008	0.968	291
	Negative	0.011	0.958	291
<i>Panel D: Alternative Pre-Publication Baseline</i>				
AFA conference baseline (+ year trend)	Directional	0.327	0.241	151
	Directional	0.345	0.220	151
Uniform LLM coding [AFA, + year]	Directional	0.352	0.204	151
	Directional	0.364	0.193	151

Notes. Each row reports the probit (or OLS for the LPM specification) coefficient on the indicated variable. The omitted category is null/mixed ($d_i = 0$), except for binary directional specifications where it is null/mixed vs. any directional finding. All probit specifications include a linear year trend unless otherwise noted. “Matched pairs only” uses the 16 papers confirmed to exist in both NBER and published form; the LLM codes each paper’s direction from its NBER *draft* abstract and from its published abstract separately, yielding $N = 32$ observations (16 draft, 16 published); all directional-finding pairs are published in this small subsample (perfect separation), so the probit cannot be identified. “Binary + year + authors” adds $\log(1 + N_{\text{authors}})$ to the binary specification. The quasi-experimental restriction uses identification strategies classified by our LLM pipeline from abstracts ($N = 139$). The abstract-length specification adds $\log(\text{abstract length})$ as a covariate; both direction coefficients remain negligible, confirming the null result is not driven by abstract-length differences. The AFA conference baseline uses AFA annual meeting papers (2006–2024, $N = 151$ in-scope) as the pre-publication comparison group; direction codes follow the same three-tier hierarchy as the NBER pipeline. “Uniform LLM coding” replaces keyword-based direction codes for unmatched AFA papers with the same LLM pipeline used for the main NBER/published sample; $\beta \approx 0.35$ remains insignificant ($N = 151$), confirming the AFA stage-identification result is not an artifact of the mixed coding method. Standard errors omitted for brevity. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7. Discussion

7.1 Policy Implications

Our results are *inconsistent with* the hypothesis that the published record of government-intervention research overstates the proportion of interventions with detectable effects due to significance-sorting. Publication rates across finding directions are nearly identical ($\approx 46\%$), and the direction of a paper’s finding has no statistically detectable association with its likelihood of appearing in a top-three finance journal. The published distribution closely mirrors the pre-publication distribution on this dimension.

To the extent that structural bias-correction methods (Andrews and Kasy, 2019) applied to specific intervention literatures (capital requirements, short-sale bans, mandatory disclosure) would detect bias, that bias does not appear to originate from direction-based editorial selection in the top-three finance journals. Regulators who rely on the *Journal of Finance*, *Review of Financial Studies*, or *Journal of Financial Economics* literature on government intervention appear to face a record that is not systematically distorted in favor of or against directional findings relative to null results.

The robust positive finding—that quasi-experimental identification quality predicts top-journal publication—has a different policy implication: the published literature is methodologically selective. Papers using RDD, IV, DiD, or event-study designs are substantially more likely to appear in print, suggesting that the published evidence base is biased toward cleaner causal identification. For regulators, this is plausibly a desirable form of selection rather than a distortion.

7.2 Comparison to Other Domains

Our $\approx 1.0\times$ directional-to-null publication ratio stands in contrast to analogous estimates from adjacent literatures. Brodeur et al. (2016) document excess mass just below the conventional 5% significance threshold ($|z| \in (1.645, 1.96)$) in the *American Economic*

Review, the *Journal of Political Economy*, and the *Quarterly Journal of Economics*, with implied ratios of roughly $1.5\text{--}2.0\times$ for borderline-significant versus just-significant findings. Franco et al. (2014), studying NSF-funded social science experiments with pre-registered designs, find that 65% of null results were never published versus 21% of significant results—also implying a roughly $3\times$ publication premium. DellaVigna and Linos (2022) find, in a sample of nudge-unit RCTs with institutional outcome tracking, that studies with detectable effects are substantially more likely to appear in print than null-result studies. Brodeur et al. (2023) document that pre-registration and pre-analysis plans reduce but do not eliminate these biases in economics.

Our published-sample caliper ratio of 1.20 is below Brodeur et al.’s estimates of $1.5\text{--}2.0\times$ for the AER/JPE/QJE, and our NBER benchmark ratio of 1.01 confirms that this domain’s research base does not itself generate excess bunching before the editorial stage. On the cross-sectional direction-sorting dimension, our $\approx 1.0\times$ directional-to-null publication ratio is well below the $3\times$ estimate of Franco et al. (2014) and the analogous findings in DellaVigna and Linos (2022). Together these comparisons suggest that the finance-regulation literature does not exhibit the significance-sorting patterns documented in other social science domains. This could reflect genuine differences in editorial norms at top finance journals, or could reflect the nature of our NBER comparison group (which captures well-resourced researchers whose null results may circulate widely, reducing the selection gap relative to the full population of research). Whether the absence of bias in top finance journals extends to lower-tier journals or comparable finance sub-fields without explicit policy content (e.g., corporate governance, asset pricing) is an important question for future research.

Within finance specifically, our within-paper caliper finding (published ratio = 1.20; NBER ratio = 1.01) can be read alongside Harvey and Liu (2021) and Chen (2021). Harvey and Liu model the same excess mass at the 5% threshold in published finance research and infer the pre-publication distribution structurally; our NBER benchmark observes it directly, confirming that the distortion is specific to the post-submission stage. Chen’s bound

shows that the *far* right tail of published t-statistics ($|t| > 4$) is too fat to be explained by p-hacking alone, implying that most published effects are genuine. Our caliper evidence is consistent: the excess mass we document is concentrated just *below* the 5% threshold ($|z| \in (1.645, 1.96)$, the 10%-significant-but-not-5% zone), while the broader distribution shows a 55–57% significance rate that is stable across published and pre-publication samples. Real effects produce the fat tail; revision-stage pressure produces the excess mass in the borderline 10%-significant zone just below 1.96. The two observations reinforce rather than contradict each other.

7.3 Mechanisms

Two analytically distinct patterns require separate mechanistic accounts. The *absence* of direction-based editorial selection—roughly equal publication rates across finding directions—and the *presence* of within-paper threshold bunching concentrated in the published record point to different stages of the research pipeline.

For the direction-sorting null result, we consider two compatible explanations below. For the caliper finding, the NBER benchmark provides a more direct answer: because the bunching is present in published papers but absent from their pre-publication counterparts, and because the overall significance rate is unchanged, the distortion is specific to the revision stage. Anti-intervention papers face elevated evidentiary demands from referees and editors—perhaps because heterodox findings invite more scrutiny—leading authors to search specifications until marginal results clear the 5% threshold. This interpretation is consistent with equal publication rates (the gate does not discriminate by direction) while generating a within-paper conditional specification-search premium for negative findings. It is also consistent with Chen (2021)’s finding that the far right tail of published t-statistics cannot be explained by p-hacking alone: the excess mass we document just below $|z| = 1.96$ reflects selective reporting of marginally significant specifications, not fabrication of large effects.

7.4 What the Data Can and Cannot Tell Us

Our empirical design estimates the cross-sectional association between finding direction and publication status. This association is consistent with two broad classes of mechanism, and distinguishing them is important because they have different policy implications. Our data cannot distinguish between them, and we do not attempt to do so.

Either mechanism could suppress a direction-publication association even if publication bias exists elsewhere in the pipeline.

Supply-side equilibrium. Researchers who anticipate that null results face no publication penalty may submit them to top journals at higher rates than in domains with known significance sorting, producing an equilibrium with equal submission rates across directions. Under this mechanism, null results are not underrepresented in the NBER pool relative to the published pool, which is precisely what we observe (NBER directional rate: 79.6%; published directional rate: 79.1%).

Identification quality as gatekeeper. The positive quasi-experimental premium (0.467^{***}) suggests that editorial selection in this domain operates primarily on methodological rigor. If clean-identification studies have directional findings at similar rates to non-causal studies in this specific domain, the identification filter would generate no direction sorting in the aggregate. Our data are consistent with this interpretation, though our design cannot confirm it.

Without submission-level data (desk-rejection rates by direction, time-to-first-decision, share of null-result papers sent out for review), we cannot test whether the absence of a direction premium in the published record reflects true editorial neutrality or an upstream equilibrium in submission behavior. Future work following the approach of DellaVigna and Linos (2022)—who obtain institutional data on submission and publication outcomes for nudge-unit RCTs—would allow a sharper test of whether journals are directionally neutral

or merely receive a directionally balanced submission pool.

7.5 Limitations

Six main threats to the validity of our analysis deserve discussion.

Fundamental identification constraint. The most important limitation is that NBER working papers and published journal articles are not different states of the same population of papers. They are largely non-overlapping samples selected through different institutional processes: only 16 of 134 published papers (11.9%) are matched to an NBER working paper. Our design therefore compares two different cross-sections, not papers tracked through the editorial filter. The observed difference in direction distributions is consistent with editorial selection but equally consistent with NBER papers having a different direction distribution for reasons unrelated to journals—for example, NBER researchers may strategically circulate papers with more ambiguous findings, or null-finding papers in this domain may disproportionately circulate through SSRN rather than NBER. Addressing this fully would require either submission-level data from journal editors (desk-rejection rates and review outcomes by direction code) or a pre-registration registry for finance papers analogous to the AEA RCT Registry. SSRN preprint histories for authors in our sample would also allow a more comprehensive pre-publication baseline by capturing papers that circulate outside NBER. We interpret all coefficients as cross-sectional associations, not causal editorial-filter effects.

LLM classification errors and abstract polish. Our pipeline assigns direction codes by reading paper abstracts. A key concern—confirmed by our data—is that NBER working paper abstracts are written in a more tentative, multi-result style than published abstracts, even though they are longer on average (158 words vs. 109 words for published abstracts). Accordingly, 81.3% of published papers receive a “high-confidence” LLM direction code, versus 62.4% of NBER-only papers. If the LLM more often assigns a null code to hedged NBER abstracts, the measured null-paper publication rate would be biased downward—which

would *attenuate* our estimated significance premium rather than inflate it.

Our replication exercise (Section 3.4) provides direct evidence on this attenuation: of the 20 null-coded papers in the validation sample, 15 are re-classified as directional in the independent replication run, confirming that the null category is the least stable. This instability *reduces* the significance premium we estimate, because some papers we code as null would be coded as directional by an alternative rater—which would increase the denominator of the directional publication rate (since the reclassified papers are unpublished NBER working papers, so the numerator is unchanged) while reducing the denominator of the null rate, inflating the measured null publication rate and compressing the estimated gap between directional and null rates. Our reported estimates should thus be interpreted as a conservative lower bound on the true bias. Future work with standardized abstract templates, full-paper coding (rather than abstract coding), or pre-registered abstract language could address this concern directly.

Quality controls and the identification-strategy alternative. The positive quasi-experimental premium (0.467***) confirms that identification quality is a genuine predictor of publication. Papers using RDD, IV, DiD, or event-study methods are approximately 18 percentage points more likely to appear in a top journal, even controlling for direction. Importantly, including this control does not alter the direction coefficients ($\hat{\beta}_1 = -0.135$, $\hat{\beta}_2 = -0.156$, both insignificant), ruling out the concern that the null result on direction reflects confounding with identification quality.

We acknowledge that the LLM-based identification-strategy coding from abstracts is itself imperfect: fewer than half of abstracts explicitly name their identification strategy, so the LLM infers it from other signals. A hand-coded quality assessment across all 291 papers—using full paper text to classify identification strategy, sample size, and data quality—would provide a more definitive quality control and is a natural extension of this work.

NBER selection and the direction of potential bias. NBER researchers are among the most productive and well-connected academics in finance. If their directional-finding

papers are more publishable than directional-finding papers from the broader population, the NBER comparison group may not be representative of the typical pre-publication paper. Under this scenario, our null result could mask publication bias present in the broader research population. We cannot rule out this possibility without submission-level data or a more representative pre-publication baseline. Absent such data, the most reassuring feature of our design is the close alignment of direction rates between NBER (79.6% directional) and published (79.1% directional), which suggests little residual direction-based sorting at least within this elite sample.

Right-censoring in the year trend. The positive and significant year trend ($\hat{\gamma} \approx 0.033^{***}$) is consistent with genuine improvement in publication prospects for recent papers, but an equally plausible mechanical explanation is right-censoring: NBER working papers from 2020–2024 have had fewer years to complete the editorial pipeline and appear in a top journal. Our cross-sectional design cannot distinguish these alternatives. The year trend should therefore be interpreted as descriptive rather than as evidence that finance journals have become more open to government-intervention research over time.

Multiple comparisons. Our main hypothesis—that directional-finding papers are published at different rates than null-finding papers—is pre-specified and tested in a single primary regression. The cross-journal and sub-period analyses are exploratory. All main results ($p > 0.90$) fall far from conventional significance thresholds; the concern is power rather than false discovery.

Our caliper analysis runs fourteen comparisons across the published sample (full sample, three journals, three directions, two tests) and mirrors them for the NBER benchmark. Applying a Bonferroni correction at the 5% level requires $p < 0.004$ for any single caliper comparison. Among the caliper tests, the full-sample published caliper ($p = 0.002$), the *Journal of Financial Economics* caliper ($p < 0.001$), and the negative-direction caliper ($p < 0.001$) all survive this threshold. Among the density-discontinuity tests, the *Journal of Finance* ($p < 0.001$) and null/mixed ($p < 0.001$) discontinuity results likewise survive

Bonferroni correction, providing independent confirmatory evidence. The corresponding NBER caliper tests ($p > 0.84$) do not approach significance, confirming the absence of pre-publication bunching. The core contrast between published bunching and NBER non-bunching is therefore robust to multiple-comparison correction.

8. Conclusion

We examine whether top finance journals exhibit publication bias in the government-intervention literature. Using 291 in-scope papers spanning 2000–2024, classified and direction-coded with an LLM-assisted pipeline, we find no evidence of direction-based sorting. Publication rates are nearly identical across positive-finding (45.6%), negative-finding (46.1%), and null-finding (46.7%) papers ($\chi^2 = 0.017$, $p = 0.992$). A binary directional probit coefficient of -0.020 (s.e. 0.182, $p = 0.91$) is statistically indistinguishable from zero and is robust across all six specifications, all three journals individually, and all three sub-periods.

The one robust predictor of top-journal publication in our sample is the use of a quasi-experimental identification strategy (RDD, IV, DiD, or event study), which carries a probit coefficient of 0.467^{***} (s.e. 0.157, $p = 0.003$). This identification-quality premium is the dominant driver of publication in our sample and is consistent with editorial reward for methodological rigor rather than for finding direction.

A complementary analysis of within-paper test statistic distributions (Brodeur et al. 2016 caliper test; 7,761 $|z|$ -statistics from 121 papers) reveals significant full-sample threshold bunching (caliper ratio = 1.20, $p = 0.002$), concentrated in *Journal of Financial Economics* papers (caliper ratio = 1.49, $p < 0.001$), while *Journal of Finance* and *Review of Financial Studies* papers do not exhibit bunching. The bunching is even more severe in negative-finding papers (caliper ratio = 1.52, $p < 0.001$)—papers that conclude interventions are harmful or ineffective—while positive-finding papers show no significant bunching.

Crucially, applying the identical pipeline to 8,351 test statistics from 125 NBER pre-

publication benchmark papers shows no bunching (full-sample ratio = 1.01, $p = 0.849$; negative-finding ratio = 1.00, $p = 0.964$), while the overall significance rate is nearly identical across the two samples (57.1% vs. 55.5%). This pre-publication benchmark localizes the distributional distortion to the revision stage rather than to the upstream research design, and is—to our knowledge—the first direct published-vs.-working-paper caliper comparison in the finance literature.

The joint pattern suggests that the *absence* of direction-based editorial selection coexists with *within-paper* threshold bunching consistent with revision-stage pressure, particularly among anti-intervention findings across journals — and concentrated in the *Journal of Financial Economics* at the journal level — though our cross-sectional design cannot rule out that this partly reflects compositional differences between NBER and published samples.

We interpret these results throughout as cross-sectional associations. Our design identifies a null association between finding direction and publication status, but cannot rule out that this reflects an upstream equilibrium in submission behavior rather than editorial neutrality. Two practical implications follow. For researchers and meta-analysts, the published distribution in this domain does not appear systematically biased toward directional findings at the editorial-selection level; concerns about cross-sectional significance-sorting in this specific corpus do not find empirical support. For journal editors, the results are consistent with a selection process that rewards identification quality over finding direction, though the within-paper test statistic patterns suggest that specification searching remains a concern at the research-design stage.

References

- Sumit Agarwal, David Lucca, Amit Seru, and Francesco Trebbi. Inconsistent regulators: Evidence from banking. *Quarterly Journal of Economics*, 129(2):889–938, 2014. doi: 10.1093/qje/qju003.
- Isaiah Andrews and Maximilian Kasy. Identification of and correction for publication bias. *American Economic Review*, 109(8):2766–2794, 2019. doi: 10.1257/aer.20171321.
- Anthropic. Claude: An AI assistant, 2024. Version claude-opus-4-7. <https://www.anthropic.com>.
- Elliott Ash and Stephen Hansen. Text algorithms in economics. *Annual Review of Economics*, 15:659–688, 2023. doi: 10.1146/annurev-economics-082222-074352.
- Allen N. Berger and Christa H. S. Bouwman. Bank liquidity creation, monetary policy, and financial crises. *Journal of Financial Stability*, 30:139–155, 2017. doi: 10.1016/j.jfs.2017.05.001.
- Utpal Bhattacharya and Hazem Daouk. The world price of insider trading. *Journal of Finance*, 57(1):75–108, 2002. doi: 10.1111/1540-6261.00416.
- Arturo Bris, William N. Goetzmann, and Ning Zhu. Efficiency and the bear: Short sales and markets around the world. *Journal of Finance*, 62(3):1029–1079, 2007. doi: 10.1111/j.1540-6261.2007.01230.x.
- Abel Brodeur, Mathias Lé, Marc Sangnier, and Yanos Zylberberg. Star wars: The empirics strike back. *American Economic Review*, 106(5):1398–1435, 2016. doi: 10.1257/aer.20150044.
- Abel Brodeur, Nikolai Cook, Jonathan Hartley, and Anthony Leigh. Mass reproducibility and replicability: A new hope, 2023. IZA Discussion Paper 16249.

- Andrew Y. Chen. The limits of p-hacking: Some thought experiments. *Journal of Finance*, 76(5):2447–2480, 2021. doi: 10.1111/jofi.13036.
- Garret Christensen and Edward Miguel. Transparency, reproducibility, and the credibility of economics research. *Journal of Economic Literature*, 56(3):920–980, 2018. doi: 10.1257/jel.20171350.
- Stefano DellaVigna and Elizabeth Linos. Rcts to scale: Comprehensive evidence from two nudge units. *Econometrica*, 90(1):81–116, 2022. doi: 10.3982/ECTA18709.
- Annie Franco, Neil Malhotra, and Gábor Simonovits. Publication bias in the social sciences: Unlocking the file drawer. *Science*, 345(6203):1502–1505, 2014. doi: 10.1126/science.1255484.
- Campbell R. Harvey and Yan Liu. Uncovering the iceberg from its tip: A model of publication bias and p-hacking. SSRN Working Paper 3865813. <https://ssrn.com/abstract=3865813>, 2021.
- Campbell R. Harvey, Yan Liu, and Heqing Zhu. ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68, 2016. doi: 10.1093/rfs/hhv059.
- Kewei Hou, Chen Xue, and Lu Zhang. Replicating anomalies. *Review of Financial Studies*, 33(5):2019–2133, 2020. doi: 10.1093/rfs/hhy131.
- John P. A. Ioannidis, T. D. Stanley, and Hristos Doucouliagos. The power of bias in economics research. *Economic Journal*, 127(605):F236–F265, 2017. doi: 10.1111/eoj.12461.
- Theis Ingerslev Jensen, Bryan T. Kelly, and Lasse Heje Pedersen. Is there a replication crisis in finance? *Journal of Finance*, 78(5):2465–2518, 2023. doi: 10.1111/jofi.13249.
- Anton Korinek. Language models and cognitive automation for economic research, 2023. NBER Working Paper 30957.

Rafael La Porta, Florencio Lopez-de Silanes, Andrei Shleifer, and Robert W. Vishny. Law and finance. *Journal of Political Economy*, 106(6):1113–1155, 1998. doi: 10.1086/250042.

J. Richard Landis and Gary G. Koch. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174, 1977. doi: 10.2307/2529310.

Appendix A. Keyword Taxonomy

Table A1 lists all terms used in Pass 1 of the classification pipeline.

Table A1. In-Scope Classification: Keyword Taxonomy

Category	Keywords
Financial regulation	regulation, regulatory, Dodd-Frank, Basel, Glass-Steagall, capital requirements, leverage ratio, bank capital
Securities law	SEC, disclosure requirement, Reg FD, Sarbanes-Oxley, SOX, insider trading law, short-selling ban, short-sale restriction
Banking policy	FDIC, deposit insurance, lender of last resort, bailout, TARP, stress test, bank resolution, government guarantee
Monetary/fiscal policy	quantitative easing, QE, Fed intervention, government subsidy, fiscal stimulus
Market microstructure rules	circuit breaker, tick size, trading halt, uptick rule, limit-down, limit-up, trading curb
Corporate governance law	board independence requirement, say-on-pay, proxy access, mandatory audit, governance mandate, clawback provision, mandatory disclosure
General intervention signals	government intervention, government policy, public policy, law and finance, legal protection, shareholder rights, creditor rights, investor protection

Ambiguous standalone terms (*regulation*, *regulatory*, *SEC*) require co-occurrence with at least one additional in-scope keyword before the paper is flagged as a candidate.

Appendix B. LLM Prompts

2.1 In-Scope Classification Prompt

A finance research paper is ‘in scope’ for a study of publication bias in government intervention research if and only if its PRIMARY research question is the CAUSAL EFFECT of a specific government law, regulation, or public policy on financial markets, firms, or investors. Exclusion criteria: (1) purely theoretical papers with no empirical test of a policy; (2) papers that only DESCRIBE a regulation without estimating its effect; (3) papers whose main topic is private firm or market behavior, with policy as incidental context; (4) papers studying central bank communication WITHOUT a specific policy instrument. Return JSON with exactly three keys: in_scope (boolean), confidence (‘‘high’’, ‘‘medium’’, or ‘‘low’’), reason (one sentence).

2.2 Direction Coding Prompt

The following is the abstract of a finance paper that studies the effect of a government law, regulation, or public policy. Classify the MAIN empirical finding as: +1 if the government intervention produced a POSITIVE or BENEFICIAL outcome (e.g., improved market quality, reduced systemic risk, better investor protection); -1 if it produced a NEGATIVE or HARMFUL outcome (e.g., reduced liquidity, higher costs, unintended consequences, welfare losses); 0 if the finding is NULL, MIXED, or INCONCLUSIVE. Coding rule: code by whether the estimated effect represents an improvement or worsening relative to the stated regulatory goal, NOT the arithmetic sign of the regression

coefficient and NOT the authors’ normative framing. Return JSON with: direction (1, -1, or 0), confidence (‘‘high’’, ‘‘medium’’, or ‘‘low’’), rationale (one sentence).

Appendix C. Additional Tables

Table A2. Probit Results by Sub-Period (*Exploratory*)

	2000–2007	2008–2015	2016–2024
Positive (+1)	0.376 (0.505) [−0.61, 1.37]	−0.247 (0.364) [−0.96, 0.47]	−0.015 (0.288) [−0.58, 0.55]
Negative (−1)	−0.537 (0.557) [−1.63, 0.56]	−0.163 (0.340) [−0.83, 0.50]	0.190 (0.274) [−0.35, 0.73]
N	50	95	146
Pseudo R^2	0.072	0.004	0.005

Notes. **This analysis is exploratory.** Sub-period sample sizes are small ($N = 50, 95, 146$) and results should be interpreted as descriptive evidence, not confirmatory tests; confidence intervals overlap substantially across periods and encompass zero in all cases. Each column estimates equation (1) on the indicated sub-period without additional controls. Standard errors in parentheses; 95% confidence intervals in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Period 2008–2015 spans the financial crisis and post-crisis regulatory response (Dodd-Frank enacted July 2010); 2016–2024 covers the post-implementation period. All direction coefficients are statistically indistinguishable from zero across all three sub-periods, consistent with the pooled null result.

Appendix D. LLM Direction Coding: Replication Confusion Matrix

Table A3. LLM Direction Coding: Replication Confusion Matrix

Original code	Replication code			Row total
	−1	0	+1	
Negative (−1)	20	0	0	20
Null/Mixed (0)	9	5	6	20
Positive (+1)	2	0	18	20
Column total	31	5	24	60

Notes. Entries are paper counts from a 60-paper stratified random subsample (5 papers per source $\times$ direction cell, balanced across the three journals and NBER). Rows show the original direction code used in the main analysis; columns show the code from an independent second LLM run (claude-haiku-4-5-20251001, temperature = 0; see Section 3.4). Bold entries on the main diagonal indicate agreement. Overall agreement: $43/60 = 71.7\%$. Linear-weighted Cohen’s $\kappa = 0.674$ (“substantial agreement” by the Landis and Koch (1977) scale). The negative category achieves 100% agreement; the positive category 90%. Disagreements are concentrated in the null/mixed category (25% agreement), with 15 of 20 null-coded papers re-classified as directional in replication—consistent with null abstracts being less decisive and more sensitive to prompt context. Because coding instability pushes ambiguous papers *toward* directional codes rather than toward null, any resulting measurement error attenuates our estimated significance premium.

Appendix E. Data and Replication

All data and code necessary to reproduce the results of this paper are available at [https://github.com/\[REPOSITORY\]](https://github.com/[REPOSITORY]). The repository contains:

- `requirements.txt` — Python package versions used
- `scripts/01_collect_nber.py` — NBER API data collection
- `scripts/02_collect_journals.py` — Journal data collection
- `scripts/02b_enrich_jfe_abstracts.py` — Elsevier API abstract enrichment for *Journal of Financial Economics*

- `scripts/03_classify_inscope.py` — Two-pass in-scope classification (keyword + LLM)
- `scripts/04_code_direction.py` — LLM direction coding
- `scripts/05_link_nber_to_published.py` — Fuzzy title matching for NBER-to-journal linkage (Round 1)
- `scripts/05b_improve_matching.py` — API-based abstract matching for unmatched NBER papers (Round 2)
- `scripts/05c_fulltext_matching.py` — LLM full-text matching via NBER PDFs (Round 3)
- `scripts/05d_published_fulltext_matching.py` — LLM matching using published PDF full text (Round 4)
- `scripts/06_analysis.main.py` — Primary probit regressions and summary statistics
- `scripts/08_robustness.py` — Robustness checks
- `scripts/09_code_id_strategy.py` — LLM identification-strategy classification (RDD, IV, DiD, event study)
- `scripts/10_collect_afa.py` — AFA annual meeting program collection (2006–2024)
- `scripts/10b_enrich_afa_abstracts.py` — Abstract enrichment for AFA papers via OpenAlex, CrossRef, and NBER
- `scripts/10c_fix_2014_2016_parsing.py` — Corrected HTML parsing for 2014–2016 AFA program pages
- `scripts/11_afa_analysis.py` — AFA-baseline probit regressions (AFA conference baseline, Section 6)
- `scripts/11c_recode_afa_direction_llm.py` — LLM direction re-coding for AFA papers (uniform coding robustness check)
- `data/` — Final analysis dataset in Parquet format
- `validation/human_coding_sheet_40.xlsx` — Coding sheet for 40-paper human direction-coding validation (to be completed before submission)

- `scripts/13_human_validation_kappa.py` — Computes Cohen’s κ between human and LLM codes once coding sheet is complete
- `paper/` — L^AT_EX source for this paper

Appendix F. LLM Pipeline Validation

Direction-coding reliability. The primary validation evidence for our LLM pipeline concerns direction coding, reported in the main text (Section 3.4). We re-ran the direction-coding protocol on a stratified random sample of 60 papers and treated the second run as an independent rater. Overall linear-weighted Cohen’s $\kappa = 0.674$ (“substantial agreement”); agreement is near-perfect for directional papers (-1 : 100%, $+1$: 90%) and concentrated in the null/mixed category (25%). The confusion matrix is reported in Appendix Table A3.

In-scope classifier design. The in-scope LLM classifier is applied to all keyword-flagged papers after a broad keyword filter selects candidates from the raw journal and NBER corpora. The keyword list is deliberately over-inclusive (any mention of regulation, law, intervention, or policy in the title or abstract), so the classifier’s primary role is removing false positives from the keyword stage rather than detecting in-scope papers from scratch.

A conservative bound on the in-scope false-positive rate is available from the data itself: all 291 in-scope papers were independently verified as on-topic during the matching and scope-review stages of the pipeline (Sections 3.3 and 3.5). No paper was flagged in-scope by the LLM and subsequently removed as out-of-scope during these downstream checks, giving a post-hoc precision of 100% within the reviewed portion of the sample.

The more material concern is the false-negative rate—papers that are genuinely in-scope but missed by the keyword filter or misclassified as out-of-scope by the LLM. False negatives would reduce the representativeness of our sample but would not mechanically bias the direction-publication association unless missed papers systematically differ in direction from included papers. Given that the keyword filter captures all standard regulatory and

supervisory terminology in finance, we expect any false negatives to be idiosyncratically missed rather than directionally selected.

Appendix G. Human Direction-Coding Validation

Table A4. Human vs. LLM Direction-Coding Agreement (N = 40)

Human code	LLM code			Total
	-1	0	+1	
-1	[-]	[-]	[-]	[-]
0	[-]	[-]	[-]	[-]
+1	[-]	[-]	[-]	[-]
Total	[-]	[-]	[-]	40

Notes. Stratified random sample of 40 papers (20 published, 20 NBER). Each paper was coded by us using only the abstract, without reference to the LLM output. Linear-weighted Cohen's κ (direction, 3-category): [-]. Cohen's κ (binary directional): [-]. Results to be completed before final submission. Human coding sheet: `validation/human_coding_sheet_40.xlsx`. Script to compute κ : `scripts/13_human_validation_kappa.py`.